Operations Research 1: Deterministic Models
Integer Programming: Solution Algorithms

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Fall, 2025

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Investments at Tatham (Revisit)

- Seven investments
  - Cash required and NPV
- Total budget: $15,000
  - Remaining cash cannot be kept.
- Objective: maximize total NPV
- What are our decisions?
  - Whether to invest in investment i, $i = 1, \ldots, 7$

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

Investments at Tatham (Revisit)

TOpS

Data

$$c_i = \text{cash required of investment } i \in I$$
$$N_i = \text{NPV of investment } i \in I$$
$$B = \text{budget}$$

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	\$5,000	\$16,000
2	\$2,500	\$8,000
3	\$3,500	\$10,000
4	\$6,000	\$19,500
5	\$7,000	\$22,000
6	\$4,500	\$12,000
7	\$3,000	\$7,500

Decisions

$$x_i = \begin{cases} 1, & \text{if we invest in investment } i \in I, \\ 0, & \text{otherwise.} \end{cases}$$

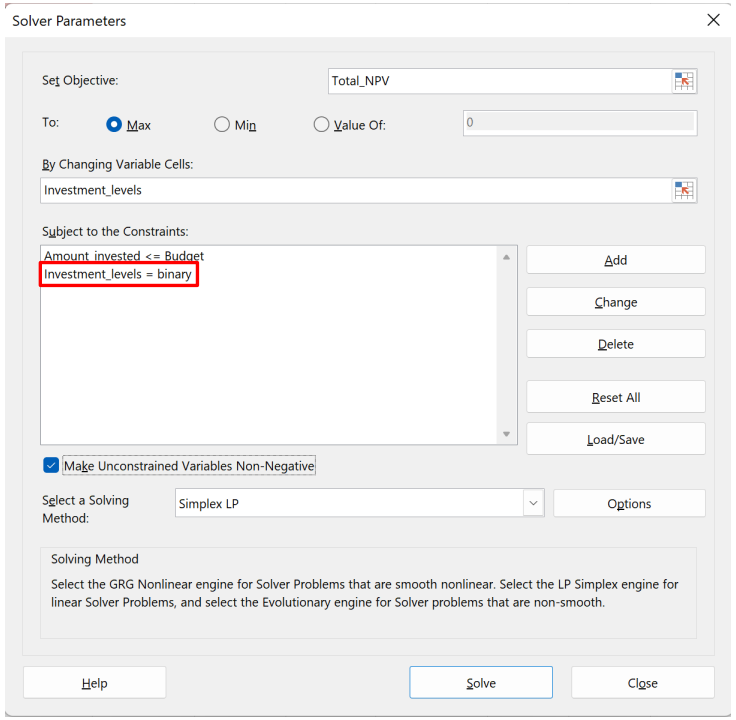
Model

$$\begin{array}{lll} \max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\ \text{s.t.} & \sum_{i \in I} c_i x_i \le B & \text{budget constraint} \\ & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0,1\}, \forall i \in I \end{array}$$

Investments at Tatham (Revisit)

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar							
Decisions: whether to invest							
Investment levels							
Budget constraint							
	Amount invested		Budget				
			$15,000				
Objective to maximize							
Total NPV							

Input data on potential investments							
Investment	1	2	3	4	5	6	7
Investment cost	$5,000	$2,500	$3,500	$6,000	$7,000	$4,500	$3,000
NPV	$16,000	$8,000	$10,000	$19,500	$22,000	$12,000	$7,500
NPV per investment dollar	3.20	3.20	2.86	3.25	3.14	2.67	2.50
Decisions: whether to invest							
Investment levels	1	1	0	0	1	0	0
Budget constraint							
	Amount invested		Budget				
	$14,500	<=	$15,000				
Objective to maximize							
Total NPV	$46,000						



IP vs. LP

IP vs. LP, Which Is (Normally) Easier to Solve?

Integer Program (IP)

- Finite number of feasible solutions
  - Exponential growth!!!
- Optimal IP solutions are NOT necessarily CPF/BF's.
- Measures of difficulties
  - Number of integer variables
  - Binary or general integer
  - Special structures
  - ……

How to Solve an IP?

How to Solve an IP?

- Example

$$
\begin{aligned}
\text{Max } & 9x_1 + 5x_2 + 6x_3 + 4x_4 \\
\text{s.t. } & \\
& 6x_1 + 3x_2 + 5x_3 + 2x_4 \le 10 \\
& x_3 + x_4 \le 1 \\
& -x_1 + x_3 \le 0 \\
& -x_2 + x_4 \le 0 \\
& x_1, x_2, x_3, x_4 \text{ binary}
\end{aligned}
$$

- IP
  - Optimal solution ($x^{IP,*}$)?
  - Optimal objective function value ($z^{IP,*}$)?

LP Relaxation

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
-x_1 + x_3 &\le 0 \\
-x_2 + x_4 &\le 0 \\
\cancel{x_1,\ x_2,\ x_3,\ x_4 \text{ binary}}
\end{aligned}
$$

$0 \le x_1,\ x_2,\ x_3,\ x_4 \le 1$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

LP Relaxation

- Example

$$
\begin{aligned}
\text{Max } & 9x_1 + 5x_2 + 6x_3 + 4x_4 \\
\text{s.t. } & \\
& 6x_1 + 3x_2 + 5x_3 + 2x_4 \le 10 \\
& x_3 + x_4 \le 1 \\
& -x_1 + x_3 \le 0 \\
& -x_2 + x_4 \le 0 \\
& \cancel{x_1,\ x_2,\ x_3,\ x_4 \text{ binary}} \\
& 0 \le x_1,\ x_2,\ x_3,\ x_4 \le 1
\end{aligned}
$$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - Question: Is $z^{LP,*}$ $(= 16.5)$ an upper bound (UB) or a lower bound (LB) of $z^{IP,*}$?

*LP Relaxation

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
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  - Question: Is $z^{LP,*}$ $(= 16.5)$ an upper bound (UB) or a lower bound (LB) of $z^{IP,*}$?
    ➔ UB

Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
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$0 \le x_1, x_2, x_3, x_4 \le 1$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - Round x_1 to an integer, i.e., $x_1 = 0$ or $x_1 = 1$?

Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
-x_1 + x_3 &\le 0 \\
-x_2 + x_4 &\le 0 \\
\cancel{x_1, x_2, x_3, x_4 \text{ binary}} & \\
0 \le x_1, x_2, x_3, x_4 &\le 1
\end{aligned}
$$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - $x_1 = 0$ ➔ Objective function value $z^{(0,1,0,1)} = 9$
  - Question: Is $z^{(0,1,0,1)}$ $(= 9)$ an upper bound (UB) or a lower bound (LB) of $z^{IP,*}$?

*Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
-x_1 + x_3 &\le 0 \\
-x_2 + x_4 &\le 0 \\
\cancel{x_1, x_2, x_3, x_4 \text{ binary}} \\
0 \le x_1, x_2, x_3, x_4 &\le 1
\end{aligned}
$$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
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  - $x_1 = 0$ ➔ Objective function value $z^{(0,1,0,1)} = 9$
  - Question: Is $z^{(0,1,0,1)}$ $(= 9)$ an upper bound (UB) or a lower bound (LB) of $z^{IP,*}$? ➔ LB

Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.
6x1 + 3x2 + 5x3 + 2x4 <= 10
x3 + x4 <= 1
-x1 + x3 <= 0
-x2 + x4 <= 0
~~x1, x2, x3, x4 binary~~

0 <= x1, x2, x3, x4 <= 1

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - $x_1 = 0$ ➔ Objective function value $z^{(0,1,0,1)} = 9$ ➔ LB
  - $x_1 = 1$?

*Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
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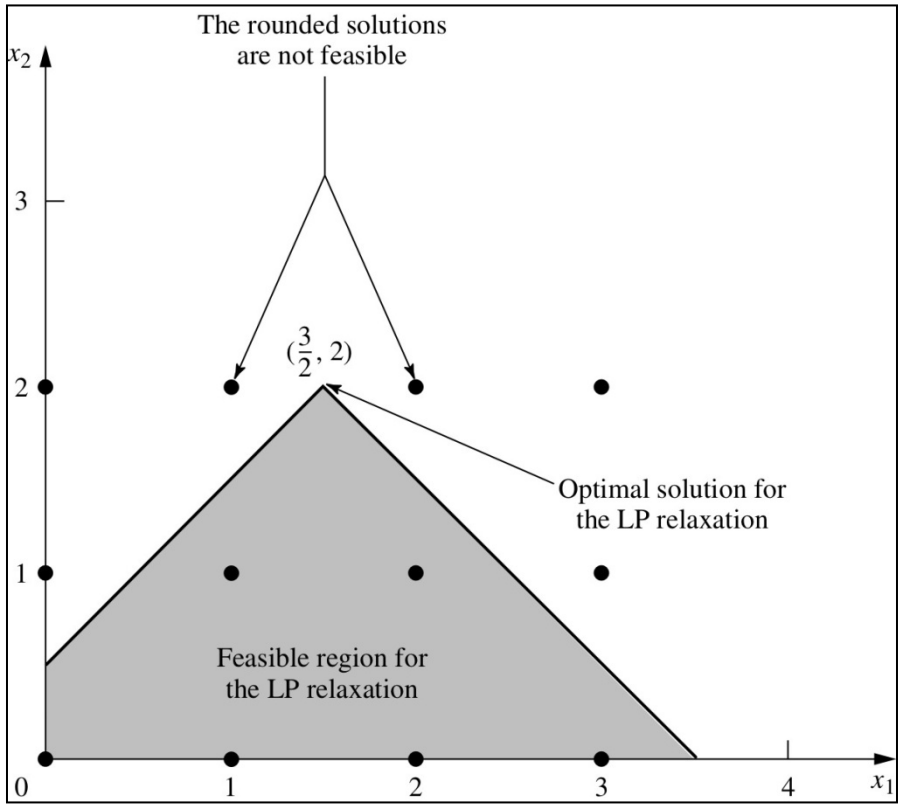
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - $x_1 = 0$ ➔ Objective function value $z^{(0,1,0,1)} = 9$ ➔ LB
  - $x_1 = 1$? ➔ Infeasible!

Round to Integer?

- May result in infeasible solutions

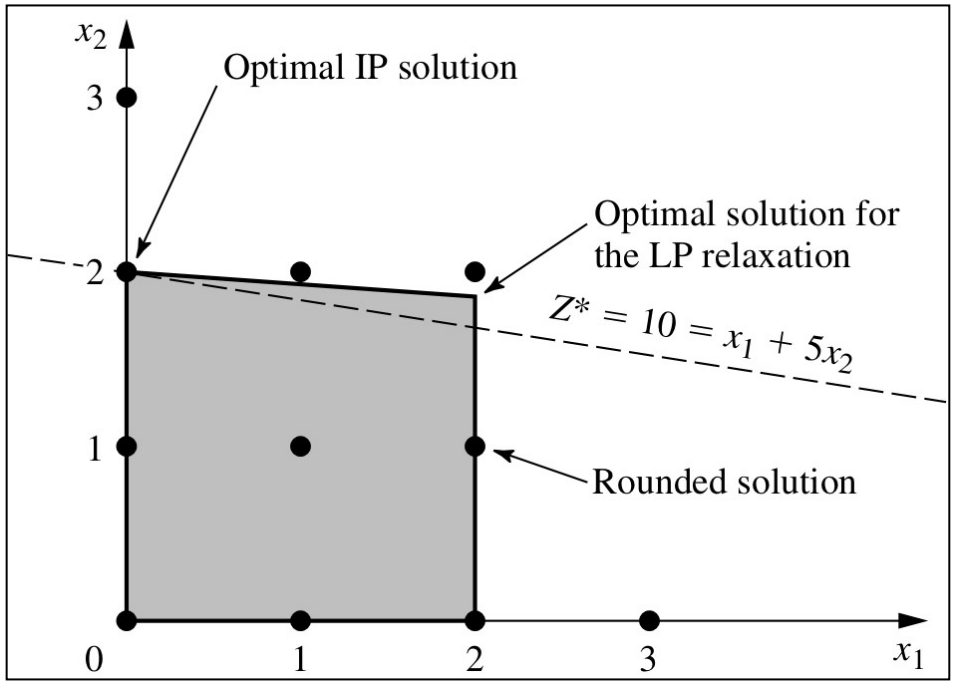
```
Max        x2
s.t.
      -x1 + x2 <= 1/2
       x1 + x2 <= 3 1/2
       x1,   x2 >= 0
       x1,   x2 int.
```



Round to Integer?

- May be far away from the optimal solution

```
Max x1 +   5x2
s.t.
      x1 + 10x2 <= 20
      x1          <= 2
      x1,      x2 >= 0
      x1,      x2 int.
```



*How to Solve an IP? (Recap)

- Example

```
Max 9x1 + 5x2 + 6x3 + 4x4
s.t.
      6x1 + 3x2 + 5x3 + 2x4 <= 10
                    x3 +   x4 <= 1
 -x1          +   x3          <= 0
        -x2          +   x4 <= 0
 x1,    x2,     x3,    x4 binary
```

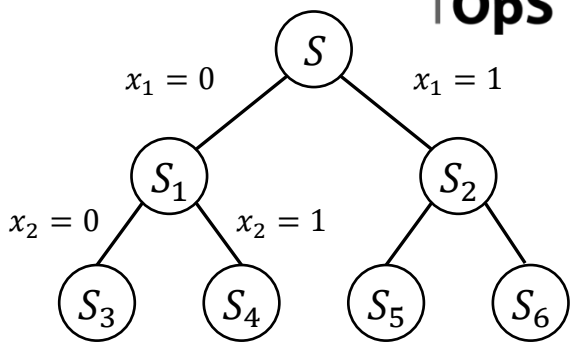
- IP
  - Optimal solution ($x^{IP,*}$)?
  - Optimal objective function value ($z^{IP,*}$)?

  - LP relaxation: $z^{LP,*}$ $(= 16.5)$ is an upper bound (UB) of $z^{IP,*}$.
  - Feasible integer solution: $z^{(0,1,0,1)}$ $(= 9)$ is a lower bound (LB) of $z^{IP,*}$.

Branch and Bound

Branch and Bound

- "Implicit" enumeration
- Divide and conquer



- Branch on non-integer solutions (e.g., $x_1 = 5/6$)
  - Depth first search strategy: find a first feasible solution
  - Best-bound first search strategy: avoid branching on any node where $\bar{z}_k \leq z^{IP,*}$
- Bound (for any node k in the branch and bound tree)
  - Solve the LP relaxation $\bar{z}_k = z_k^{LP,*} = \max\{cx \mid x \in S_k\}$ ➔ upper bound on $z_k^{IP,*}$
- Fathom/Prune at node k (for maximization problems)
  - F1: Fathom/Prune by (global lower) bound: $\bar{z}_k = z_k^{LP,*} \leq \underline{z}$
  - F2: Fathom/Prune by infeasibility: no feasible solution
  - F3: Fathom/Prune by optimality: integer solution ➔ update incumbent solution and (global) lower bound ($\underline{z}$) by $z_k^{IP,*}$

Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
-x_1 + x_3 &\le 0 \\
-x_2 + x_4 &\le 0 \\
\cancel{x_1, x_2, x_3, x_4 \text{ binary}}
\end{aligned}
$$

$0 \le x_1, x_2, x_3, x_4 \le 1$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - Branch on x_1

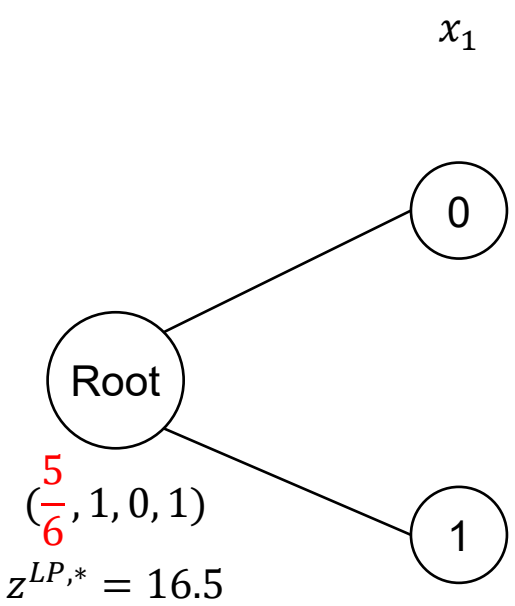
*Branch and Bound

Root

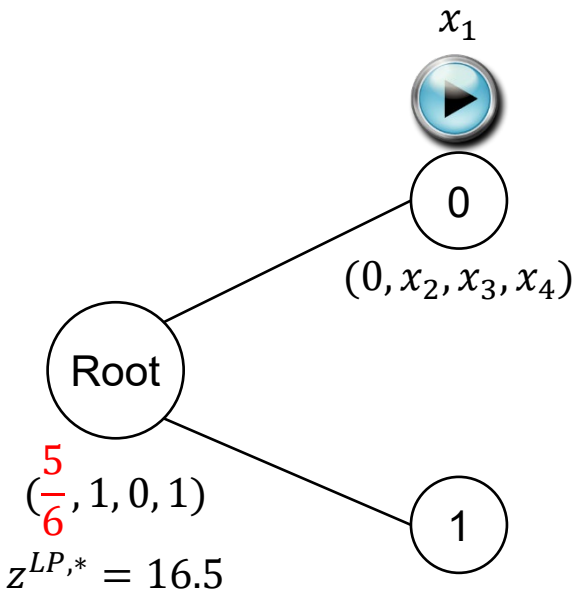
$(\frac{5}{6}, 1, 0, 1)$

$z^{LP,*} = 16.5$

*Branch and Bound

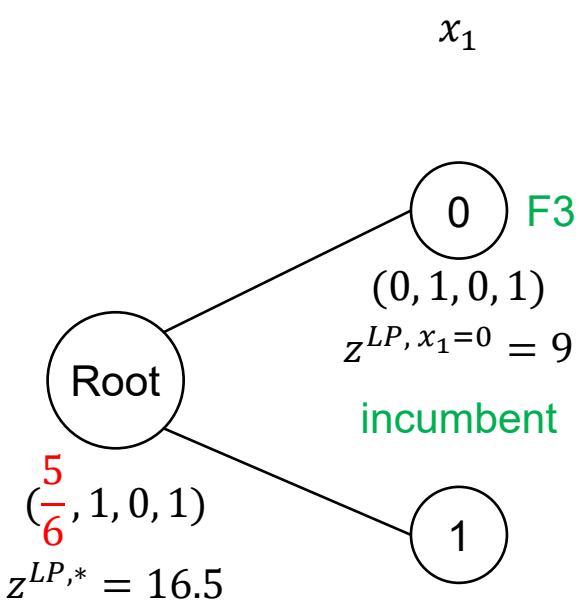


*Branch and Bound




```
! branch-and-bound example;

! branch (x1 = 0);

! objective;
max = 5*x2 + 6*x3 + 4*x4;

! constraints;
      3*x2 + 5*x3 + 2*x4 <= 10;
               x3 +   x4 <= 1;
               x3        <= 0;
      - x2         +   x4 <= 0;

! relaxation on binary integer variables;
        x2               <= 1;
               x3        <= 1;
                      x4 <= 1;
        x2               >= 0;
               x3        >= 0;
                      x4 >= 0;
end
```

*Branch and Bound



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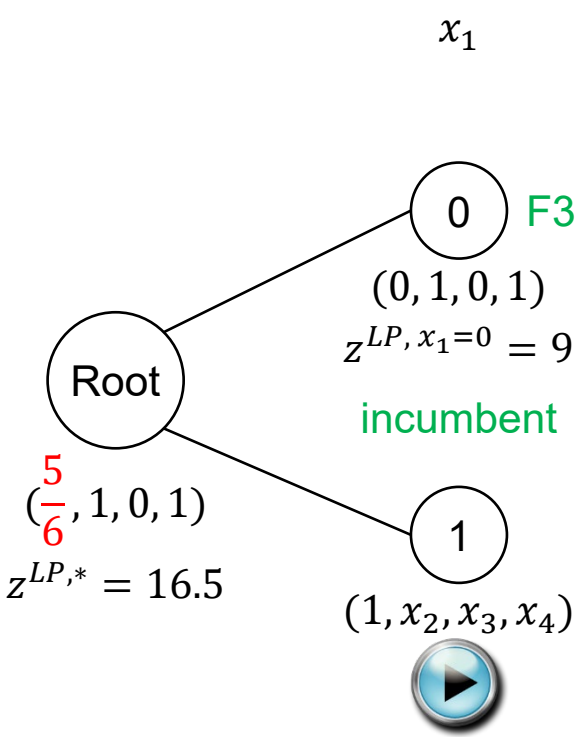


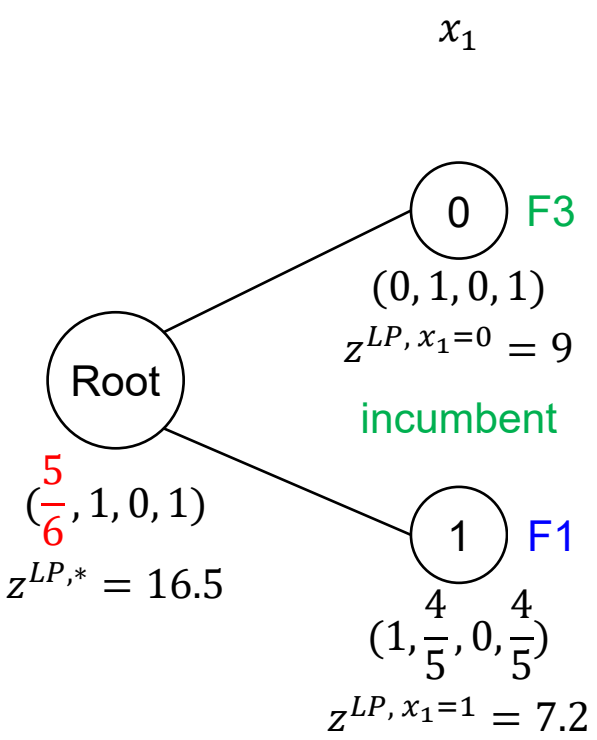

```
! branch-and-bound example;

! objective;
max = 5*x2 + 6*x3 + 4*x4;

! constraints;
      3*x2 + 5*x3 + 2*x4 <= 4;
               x3 +   x4 <= 1;
               x3        <= 1;
      - x2         +   x4 <= 0;

! relaxation on binary integer variables;
        x2               <= 1;
               x3        <= 1;
                      x4 <= 1;
        x2               >= 0;
               x3        >= 0;
                      x4 >= 0;
end
```

*Branch and Bound



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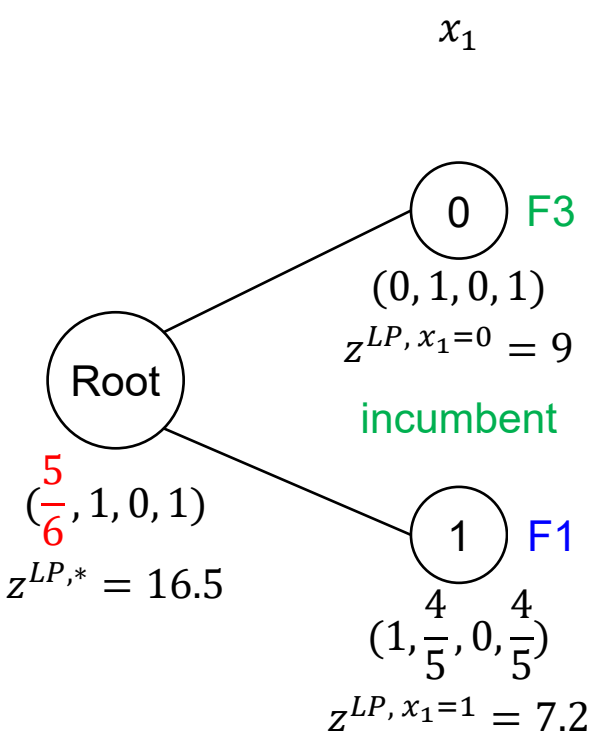

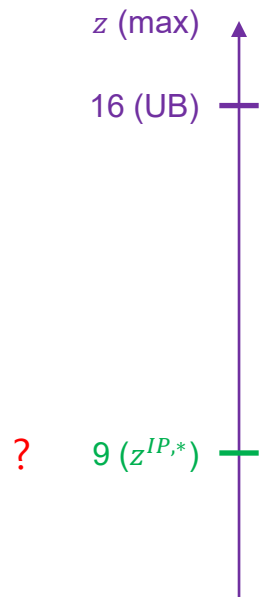

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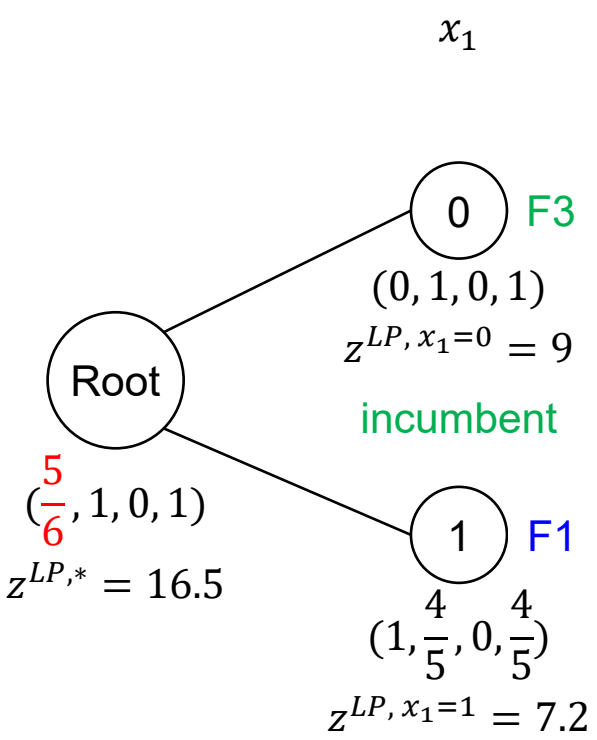

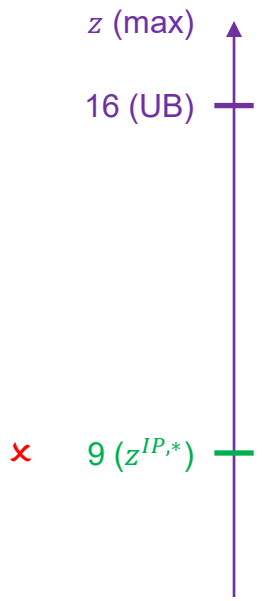

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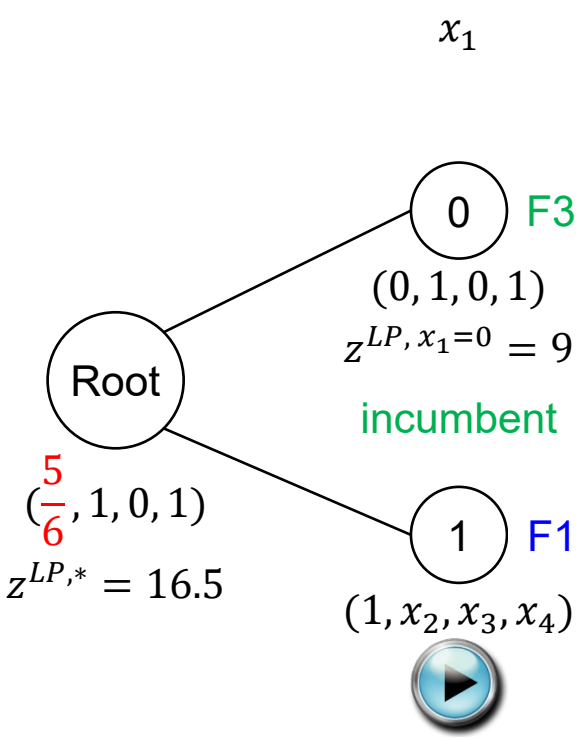
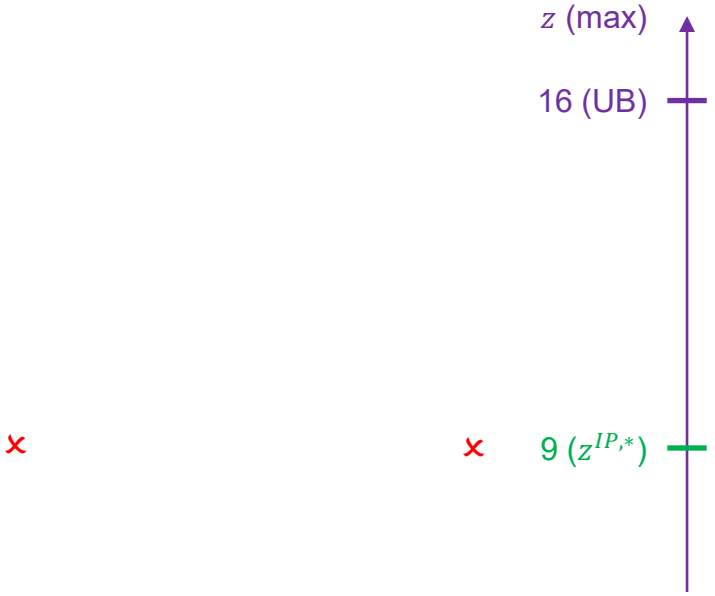
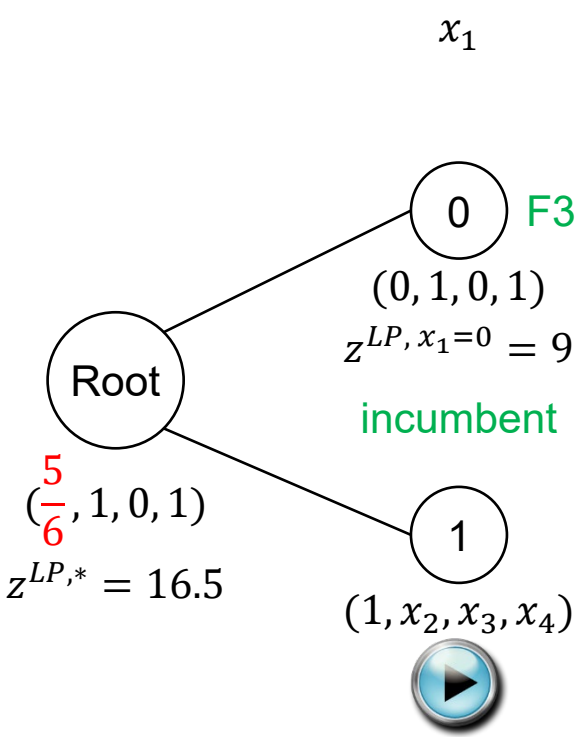


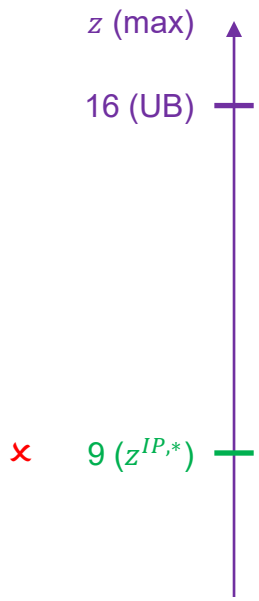

```
! branch-and-bound example;

! objective;
max = 5*x2 + 6*x3 + 4*x4;

! constraints;
      3*x2 + 5*x3 + 2*x4 <= 4;
               x3 +   x4 <= 1;
               x3        <= 1;
      - x2         +   x4 <= 0;

! relaxation on binary integer variables;
        x2               <= 1;
               x3        <= 1;
                      x4 <= 1;
        x2               >= 0;
               x3        >= 0;
                      x4 >= 0;
end
```



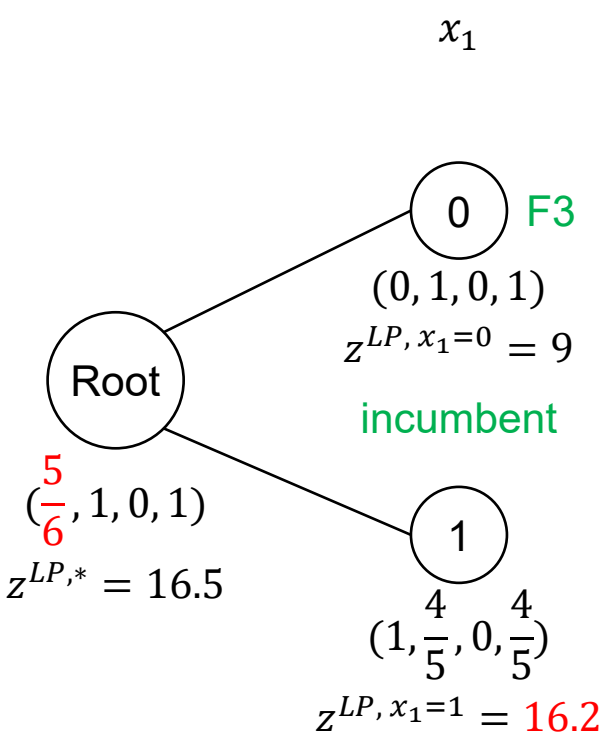
*Branch and Bound



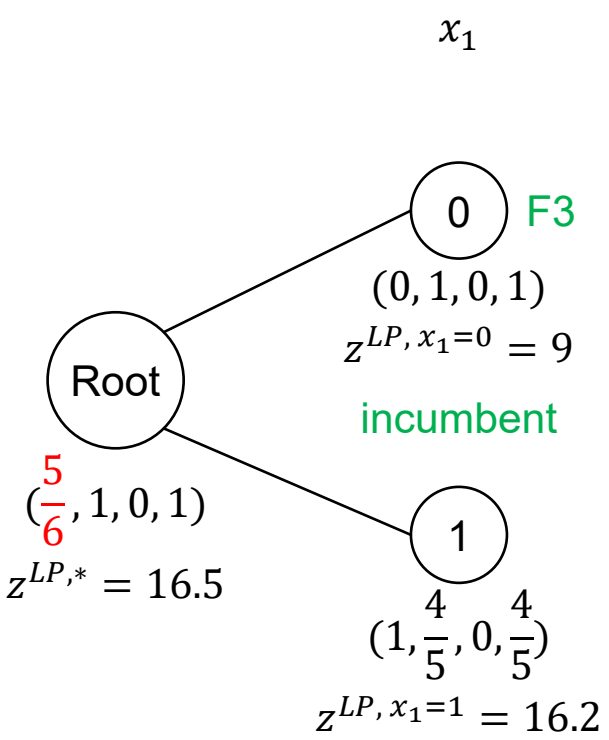

```
! branch-and-bound example;

! branch (x1 = 1);

! objective;
! max = 9*x1 + 5*x2 + 6*x3 + 4*x4;
! obj = 9*x1 + max = 9 + max;
max = 5*x2 + 6*x3 + 4*x4;

! constraints;
!     6*x1 + 3*x2 + 5*x3 + 2*x4 <= 10;
             3*x2 + 5*x3 + 2*x4 <= 4;
!                    x3 +   x4 <= 1;
                     x3 +   x4 <= 1;
!     - x1        +  x3        <= 0;
                     x3        <= 1;
!            - x2        +   x4 <= 0;
             - x2        +   x4 <= 0;

! relaxation on binary integer variables;
!       x1                     <= 1;
                x2             <= 1;
                       x3      <= 1;
                              x4 <= 1;
!       x1                     >= 0;
                x2             >= 0;
                       x3      >= 0;
                              x4 >= 0;

! @BIN(x1);
! @BIN(x2);
! @BIN(x3);
! @BIN(x4);
end
```



*Branch and Bound

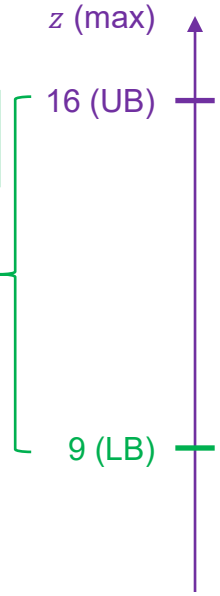


*Branch and Bound



Optimality gap
= (bound – incumbent) / incumbent

Optimality gap
= (UB – LB) / LB
= 77.78%
(for maximization problems)



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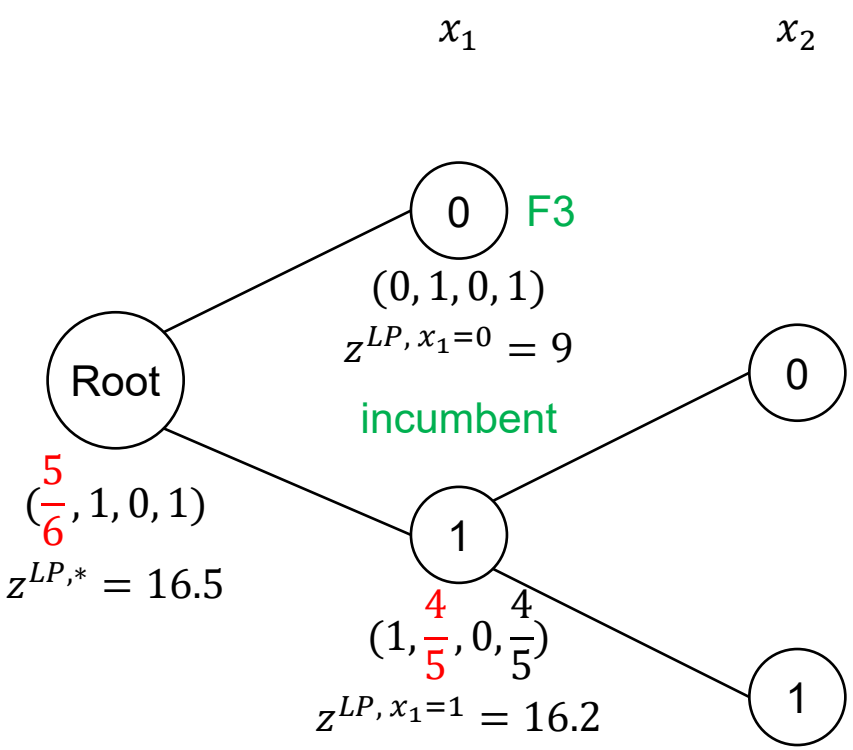

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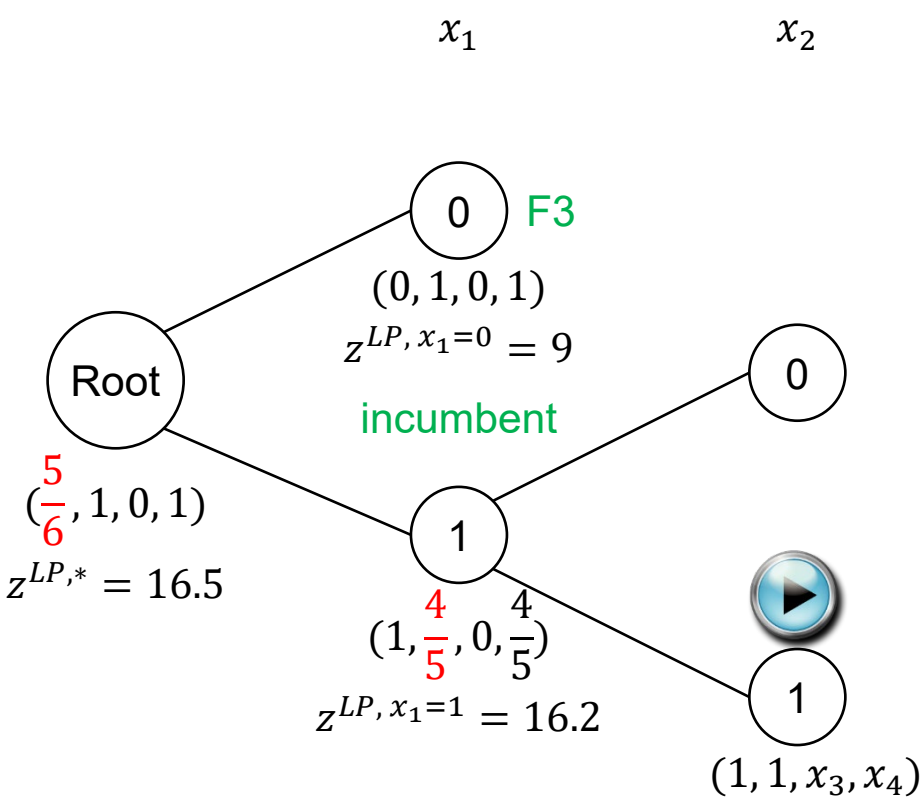
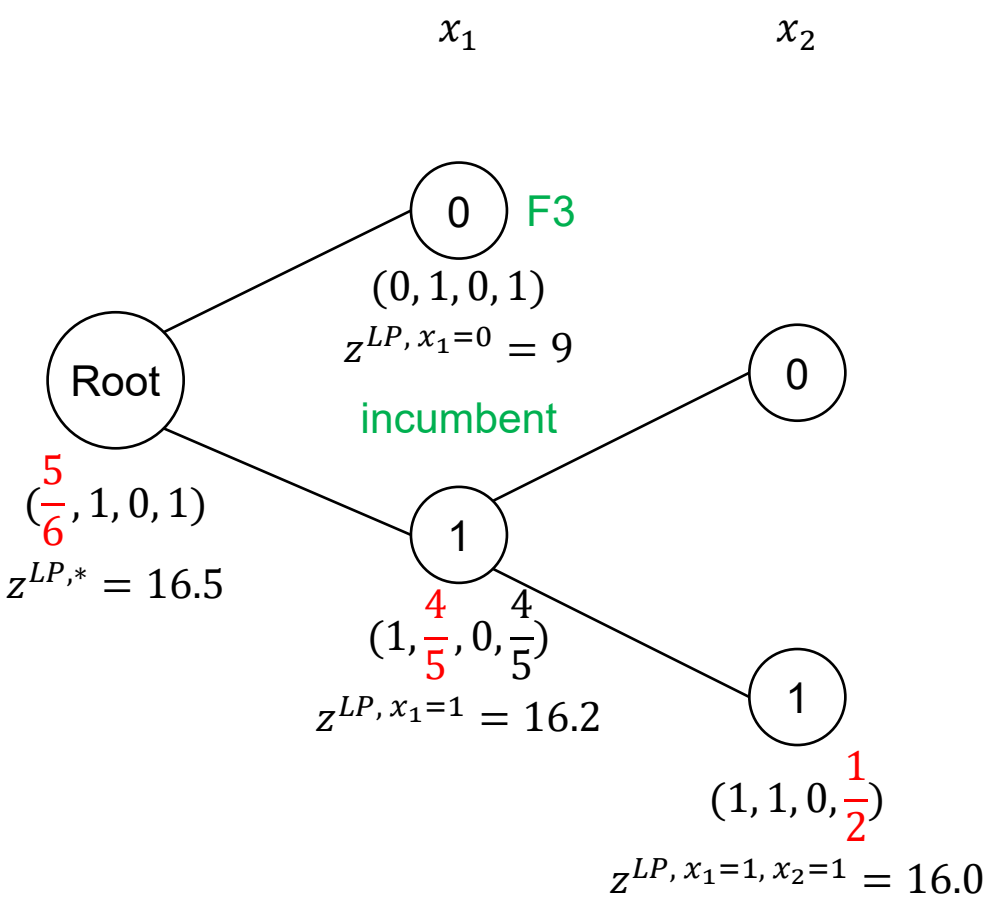



```
! branch-and-bound example;

! branch (x1 = 1, x2 = 1);

! objective;
! max = 9*x1 + 5*x2 + 6*x3 + 4*x4;
! obj = 9*x1 + 5*x2 + max = 9 + 5 + max;
max = 6*x3 + 4*x4;

! constraints;
!     6*x1 + 3*x2 + 5*x3 + 2*x4 <= 10;
                     5*x3 + 2*x4 <= 1;
!                      x3 +   x4 <= 1;
                       x3 +   x4 <= 1;
!     - x1        +    x3        <= 0;
                       x3        <= 1;
!            - x2         +   x4 <= 0;
                                x4 <= 1;

! relaxation on binary integer variables;
!       x1                      <= 1;
!              x2               <= 1;
                       x3       <= 1;
                               x4 <= 1;
!       x1                      >= 0;
!              x2               >= 0;
                       x3       >= 0;
                               x4 >= 0;

! @BIN(x1);
! @BIN(x2);
! @BIN(x3);
! @BIN(x4);
end
```

*Branch and Bound



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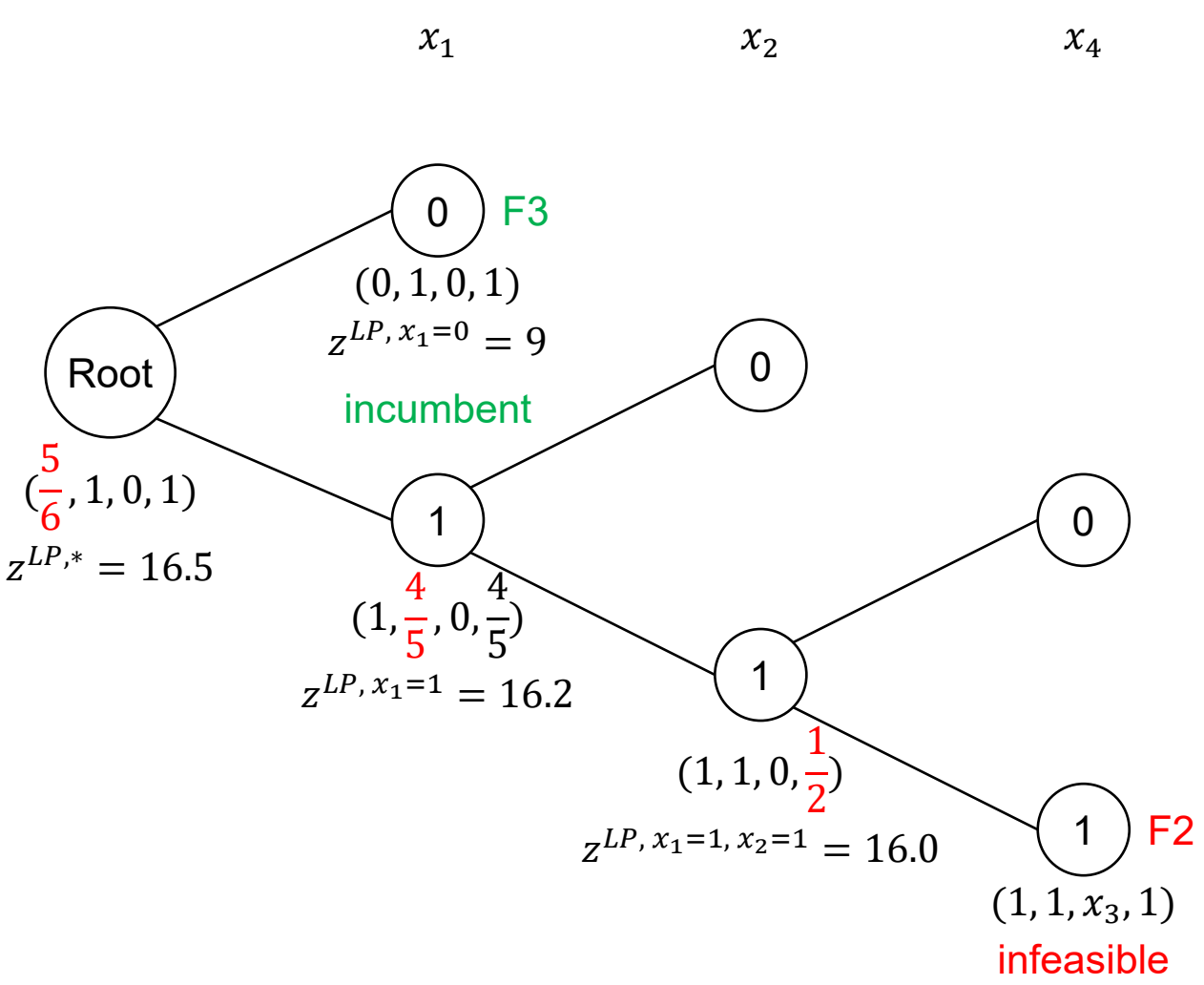

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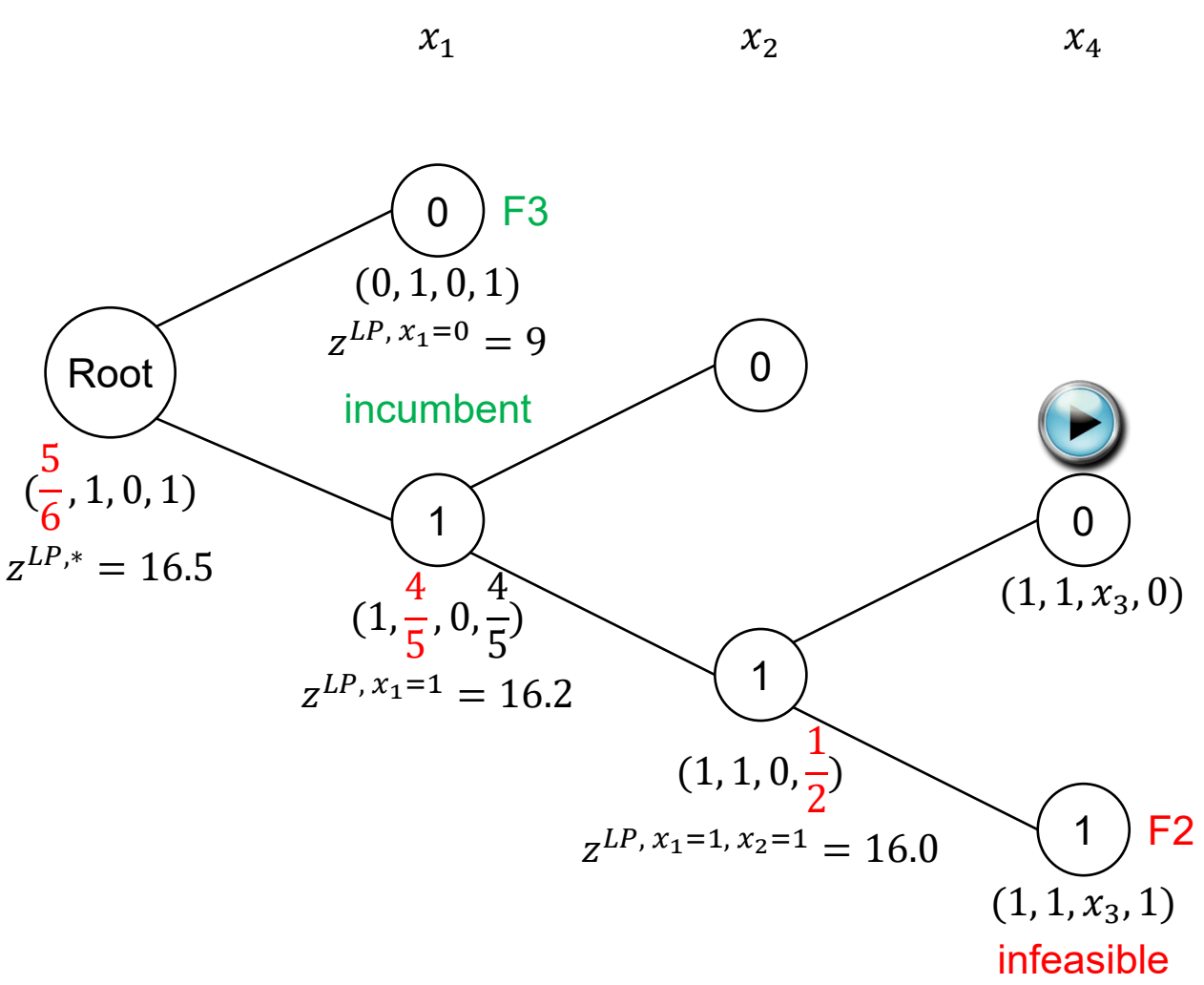

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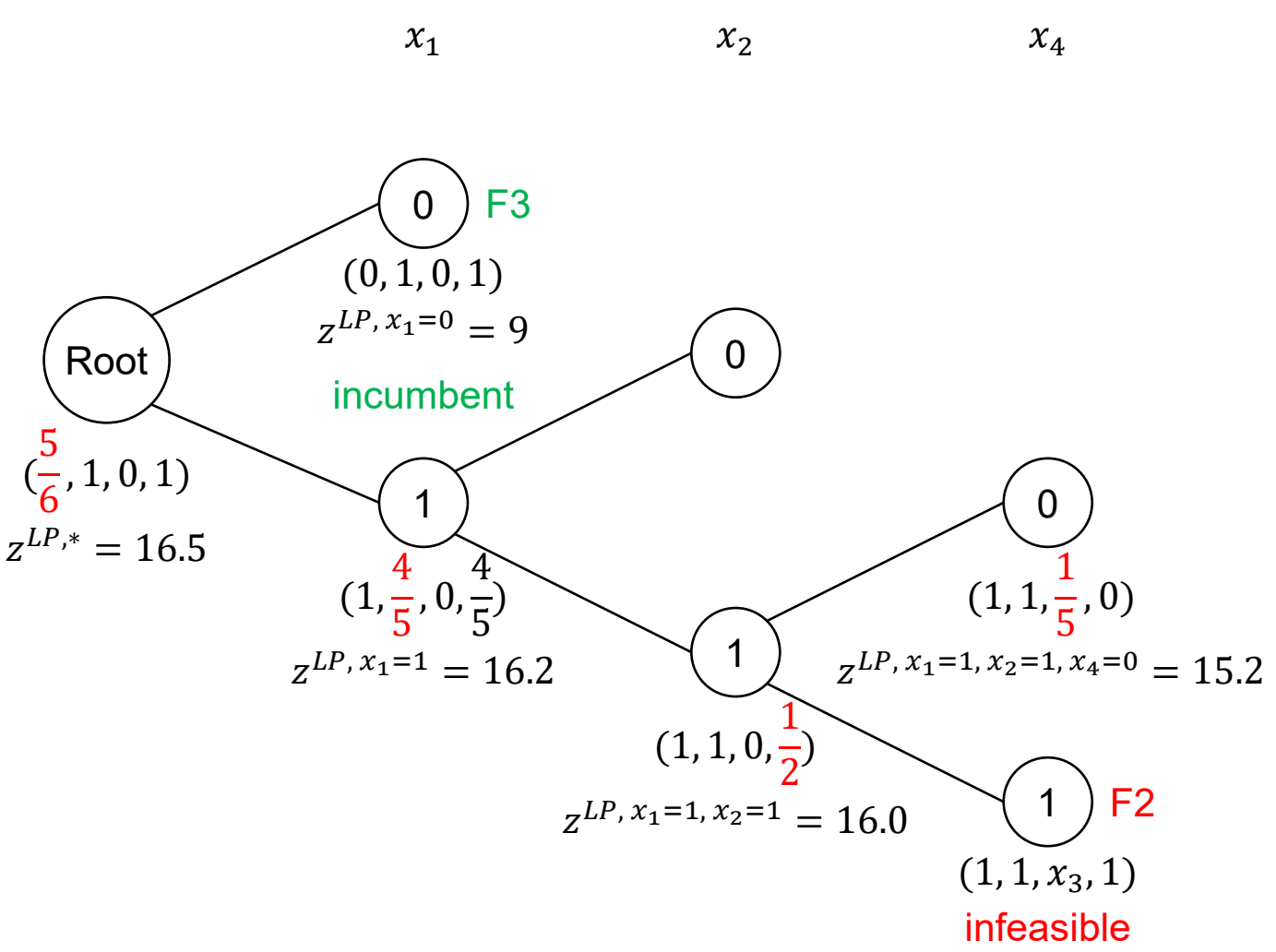

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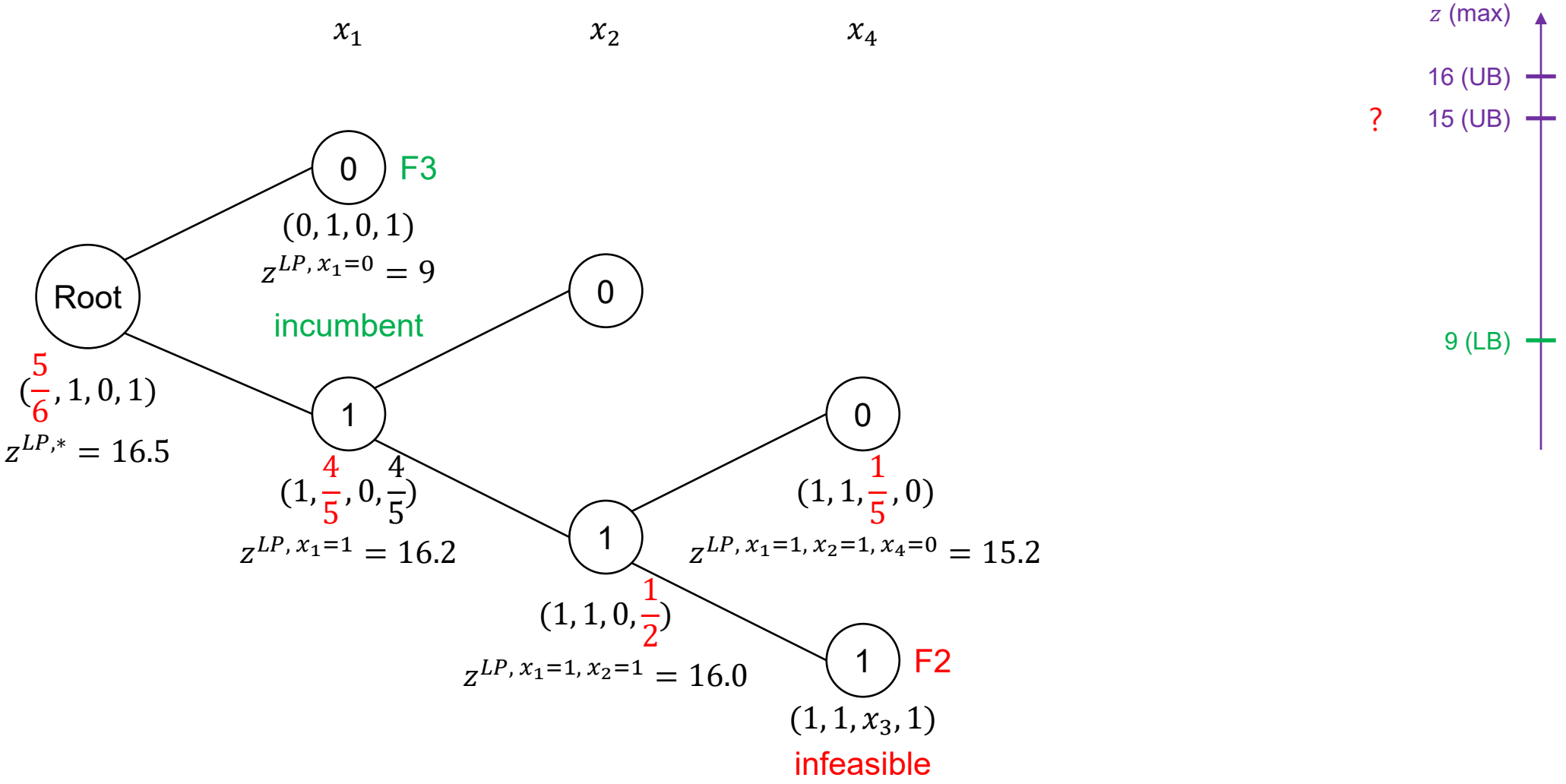

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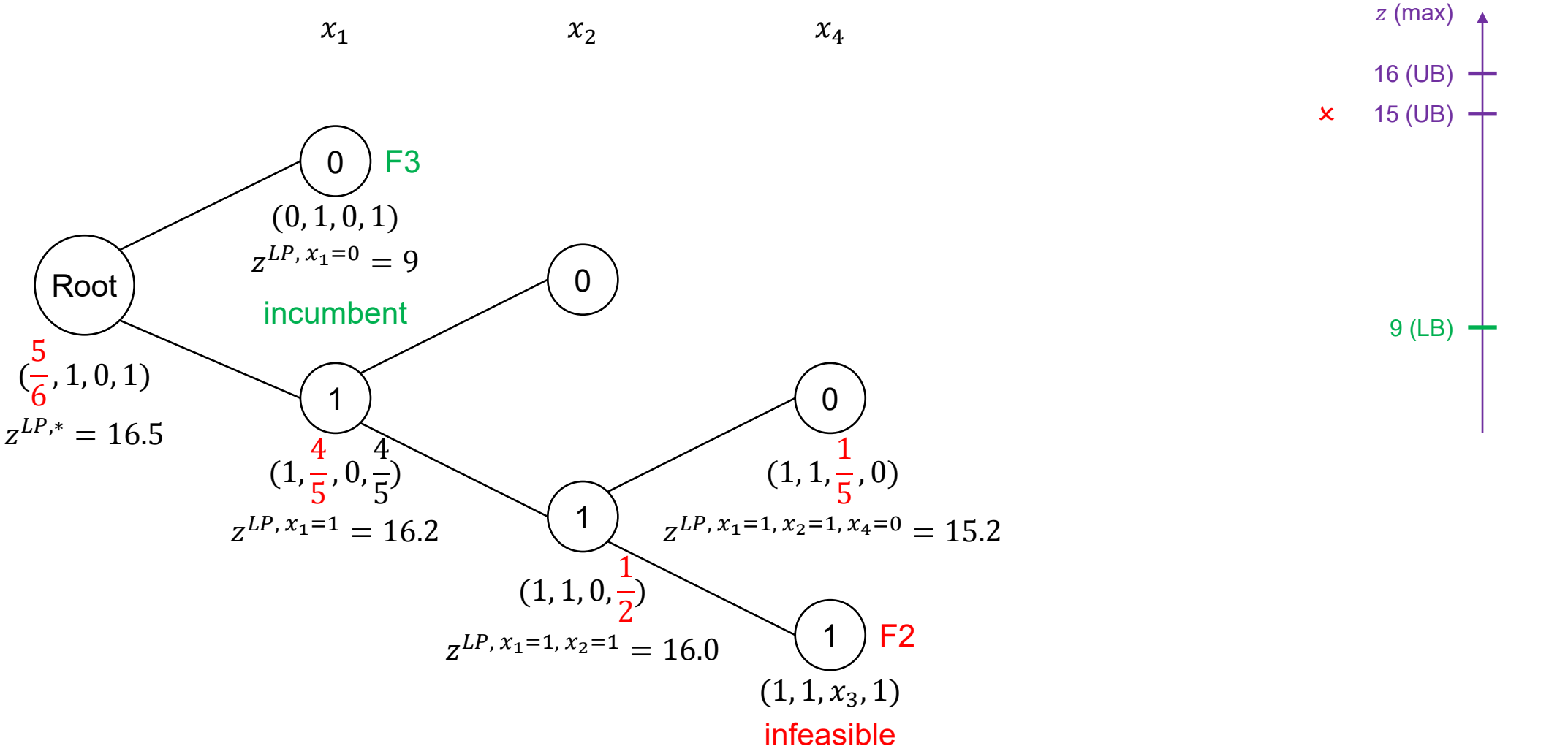

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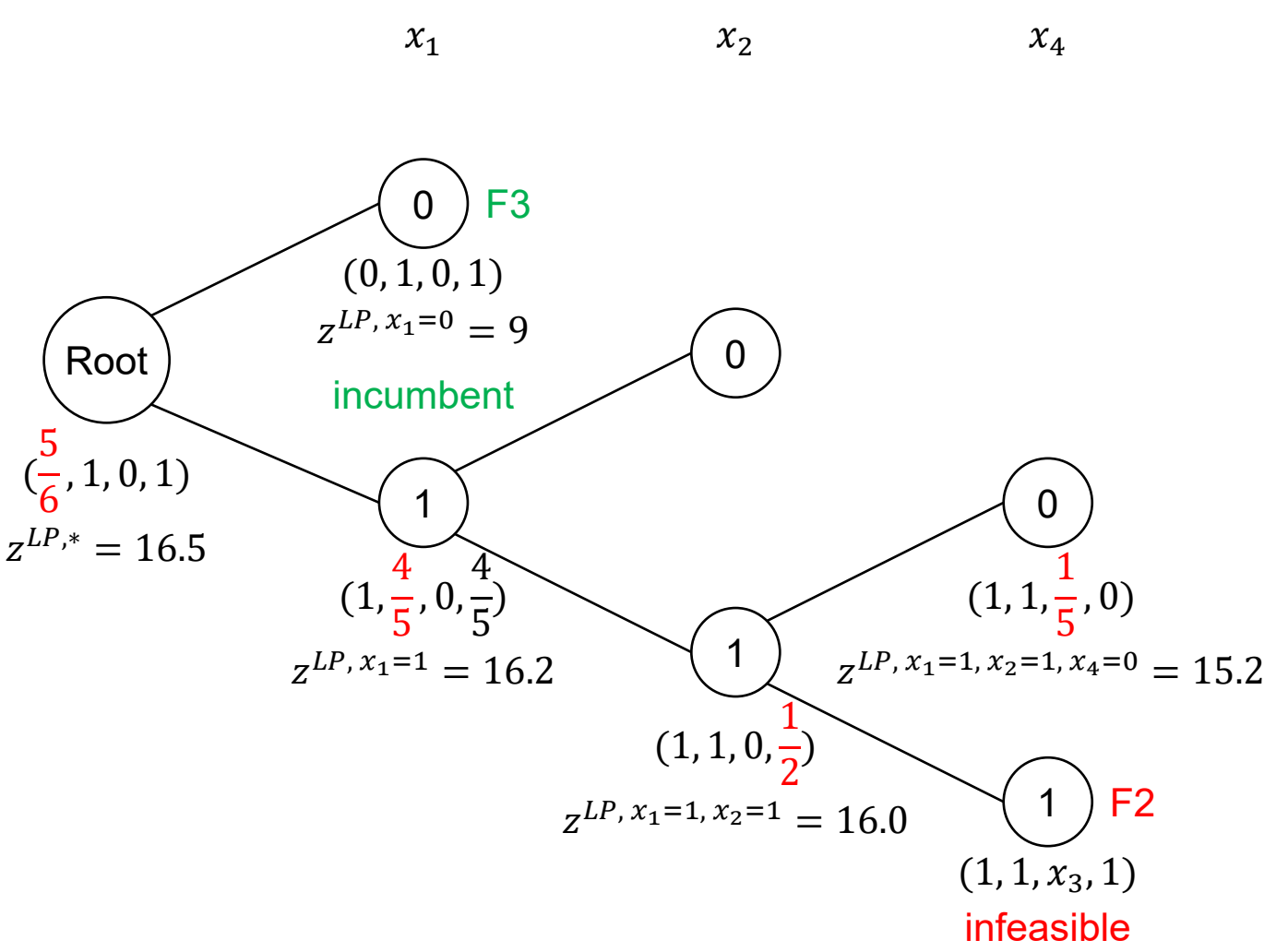

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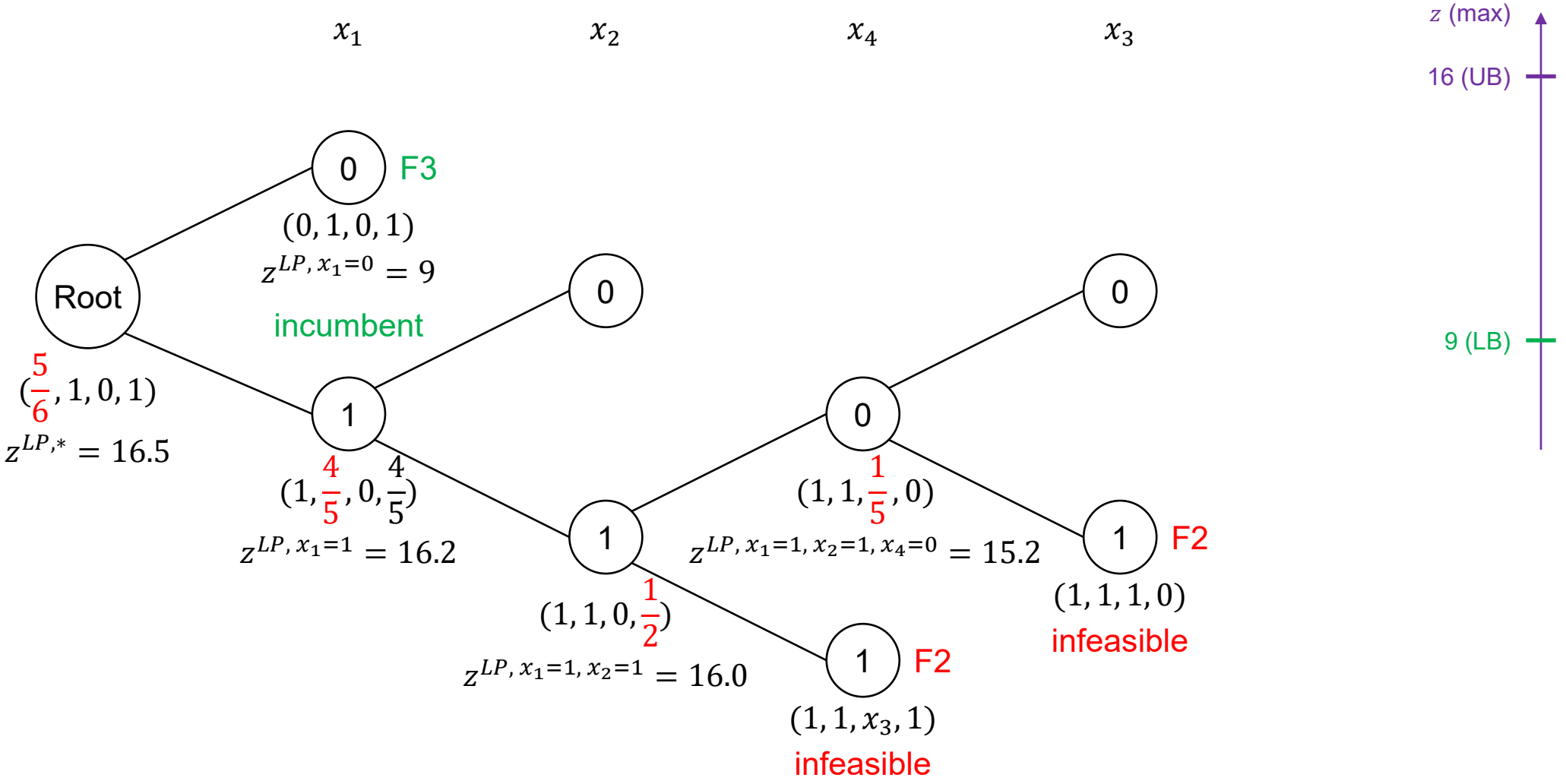

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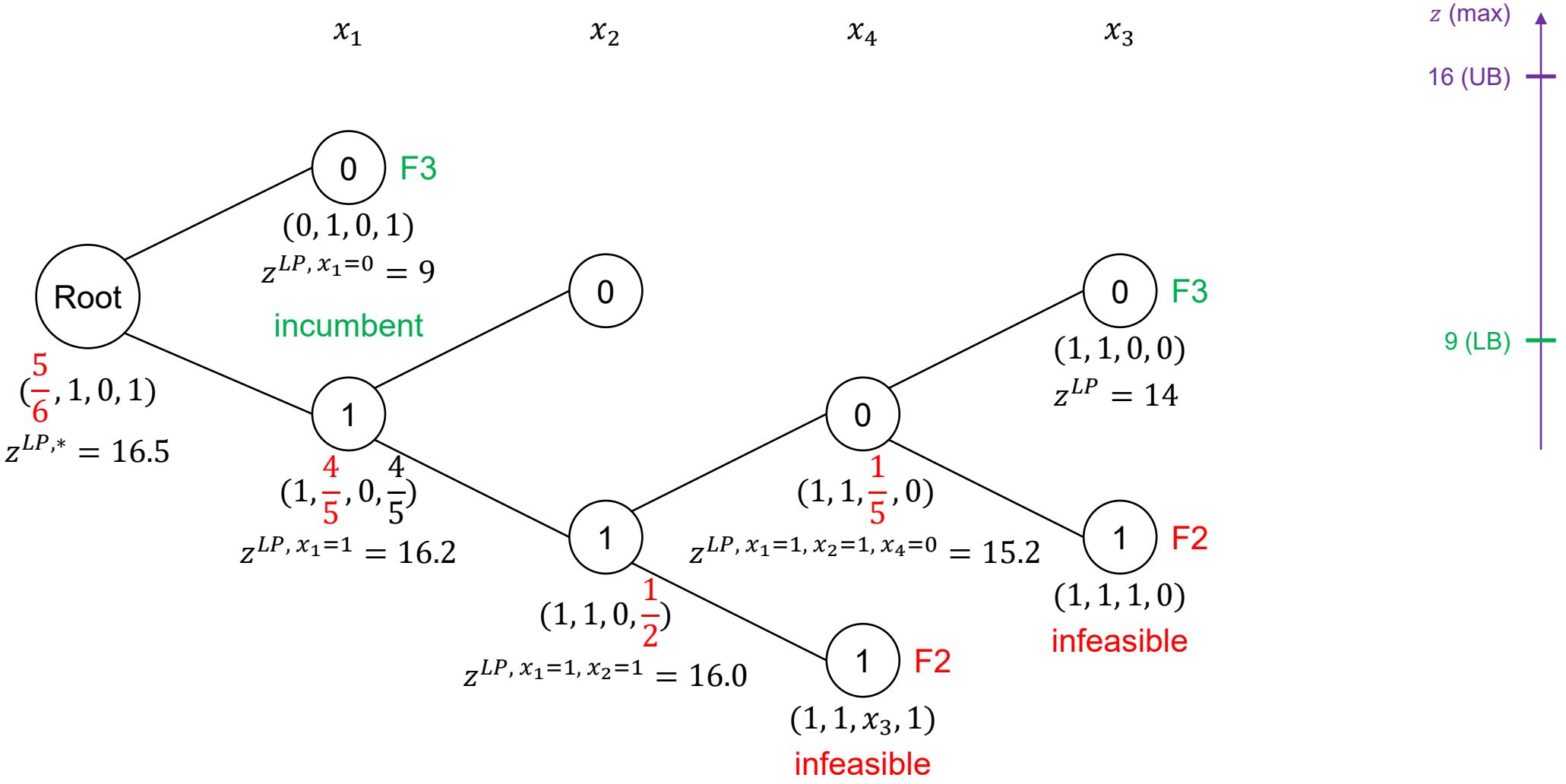

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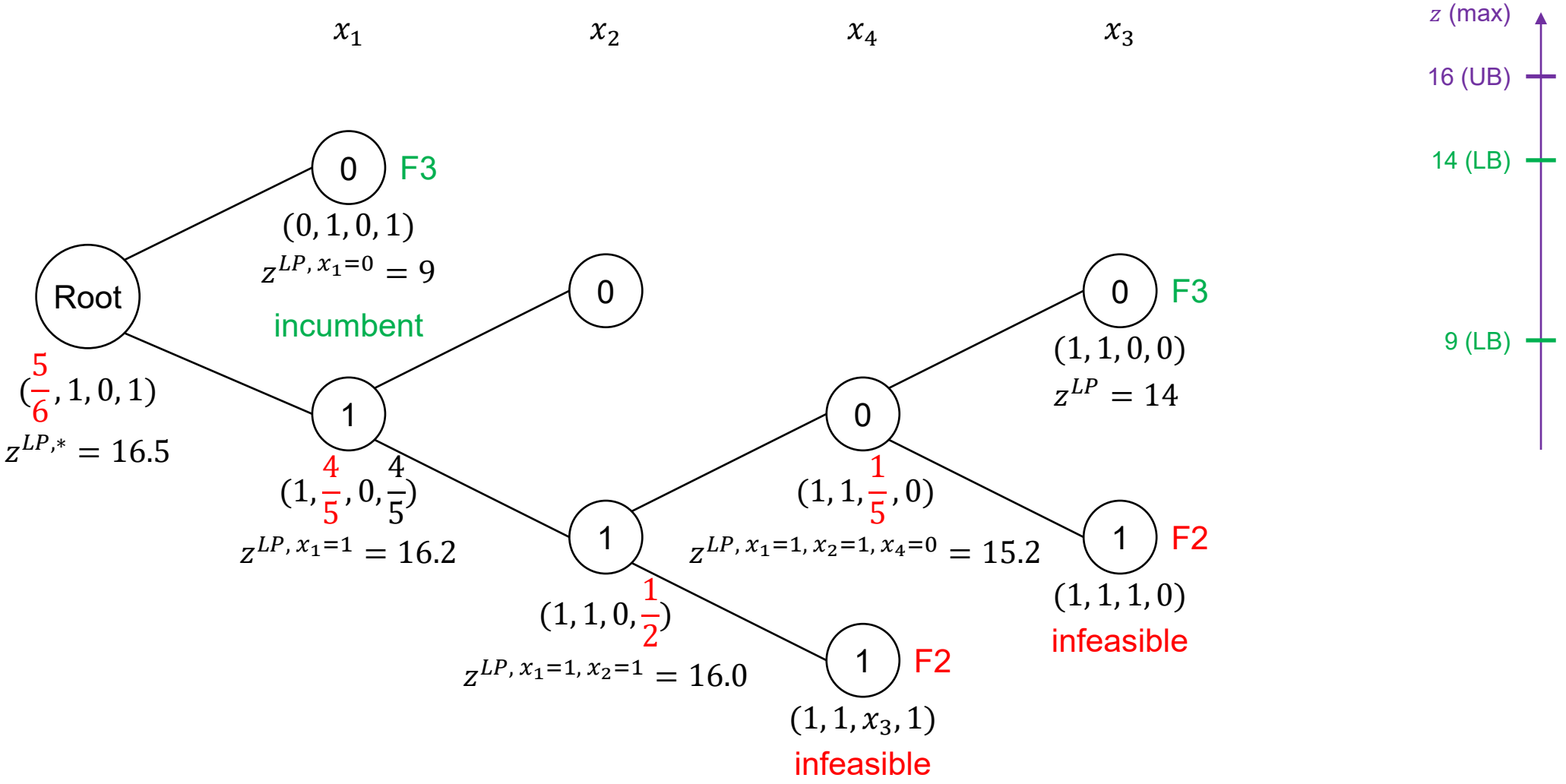

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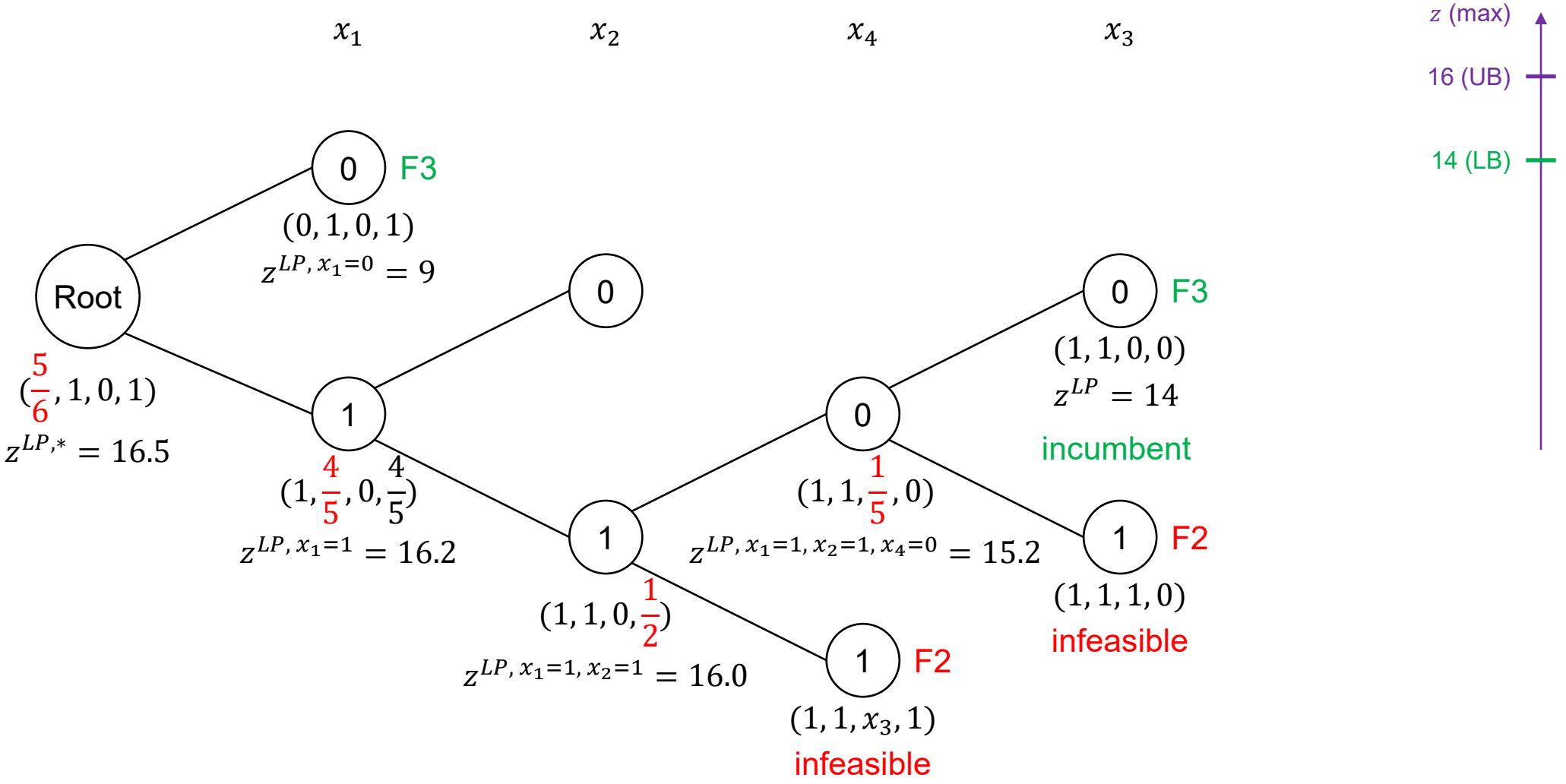

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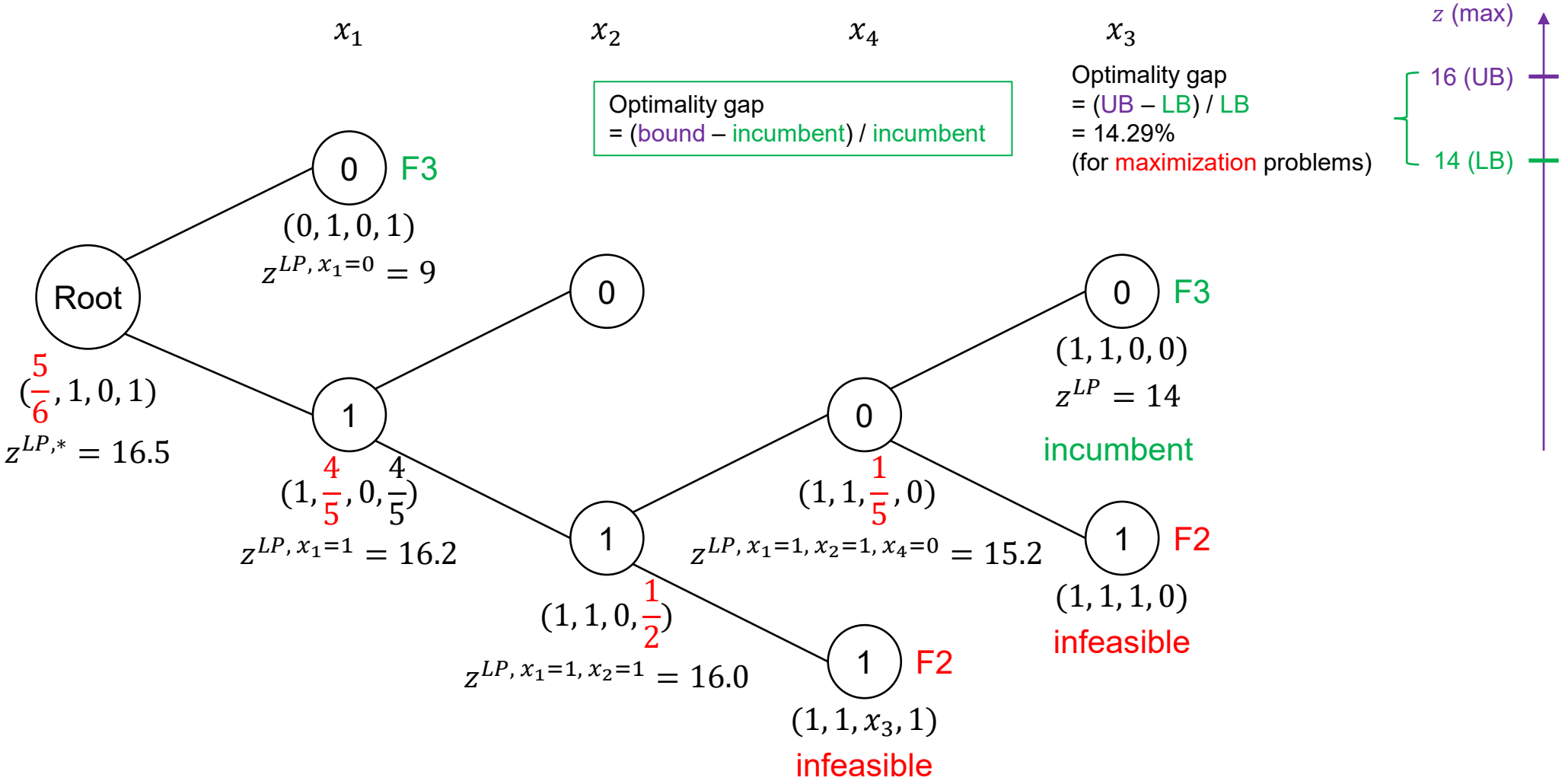

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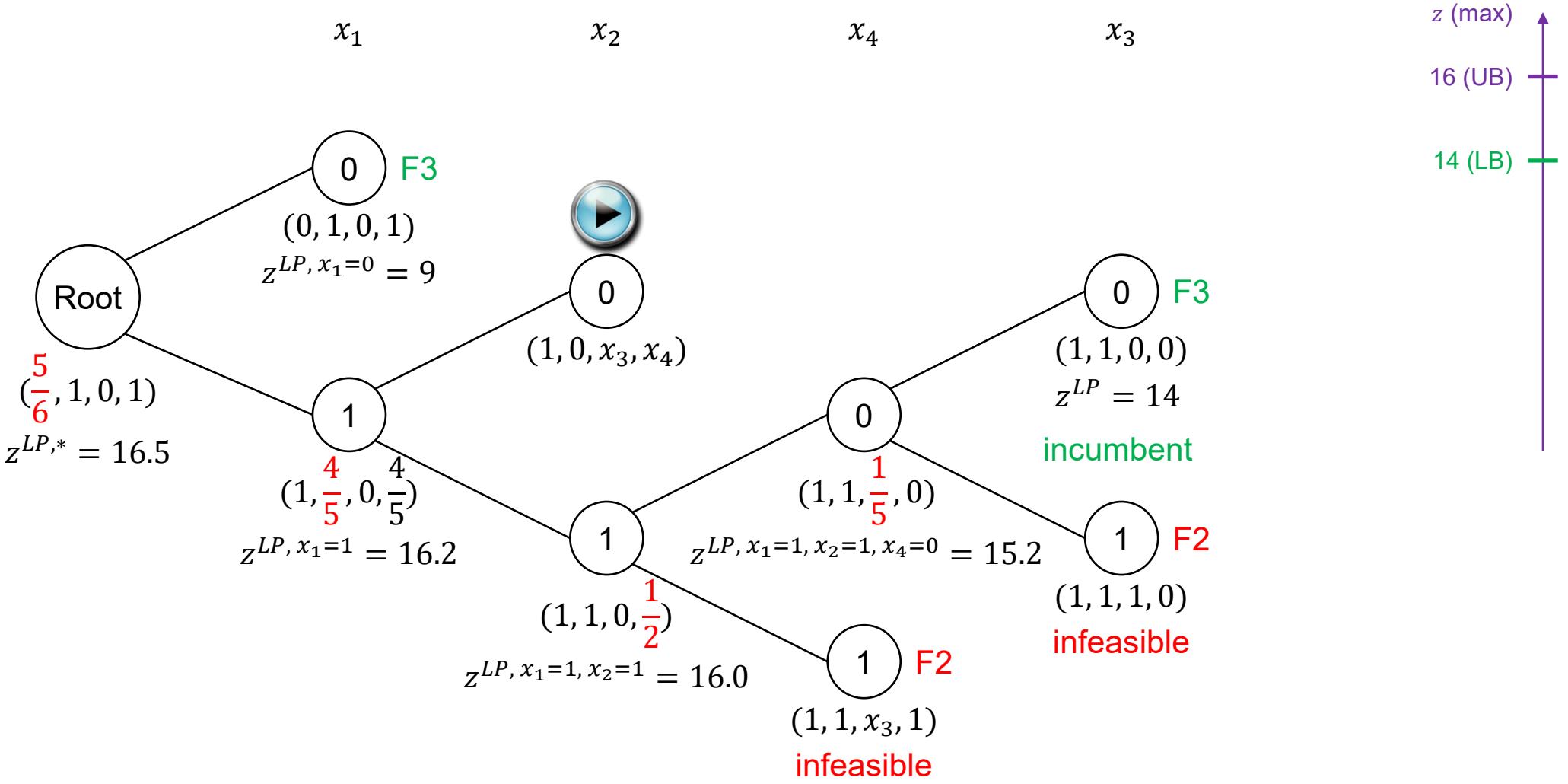

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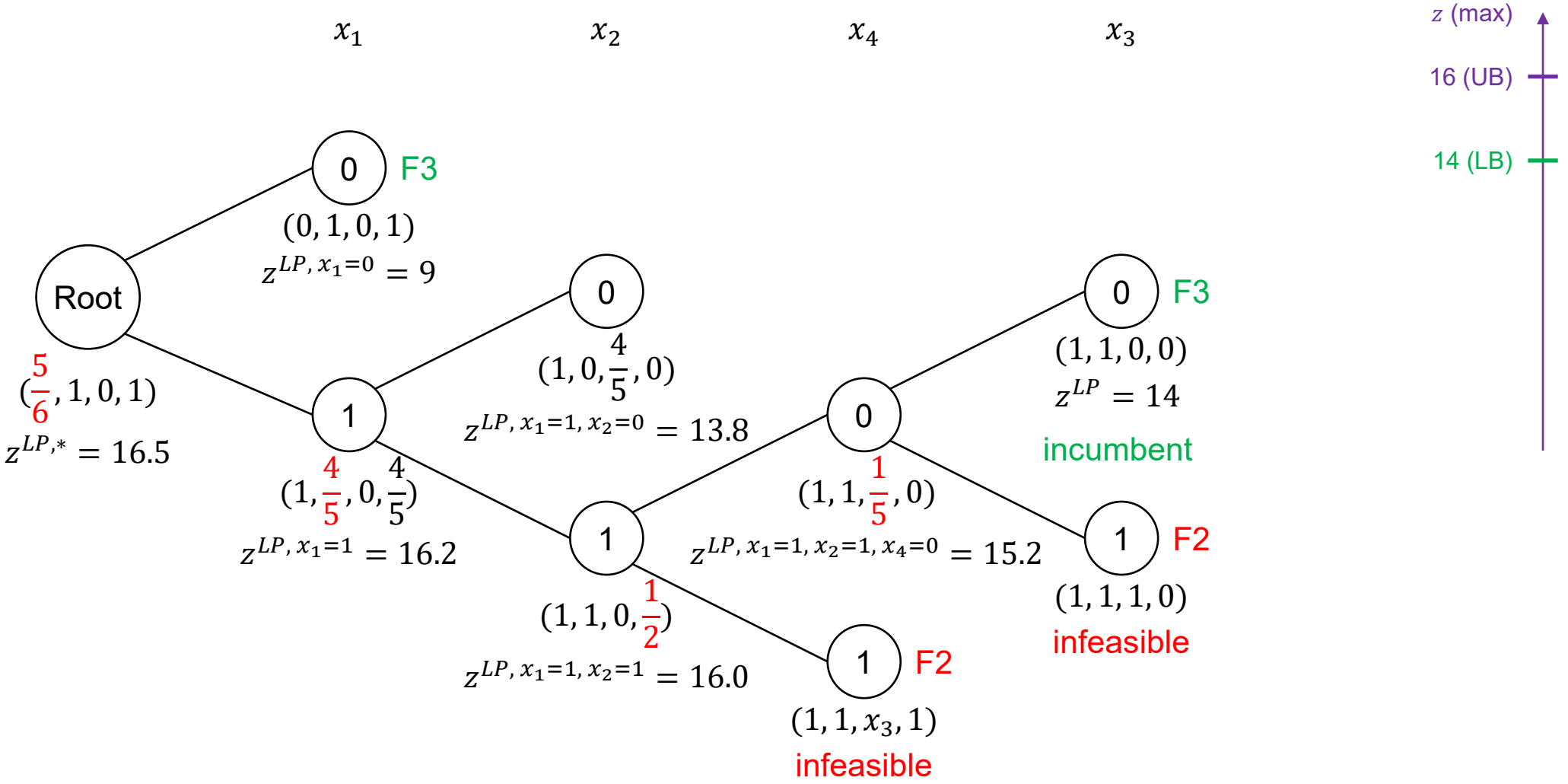

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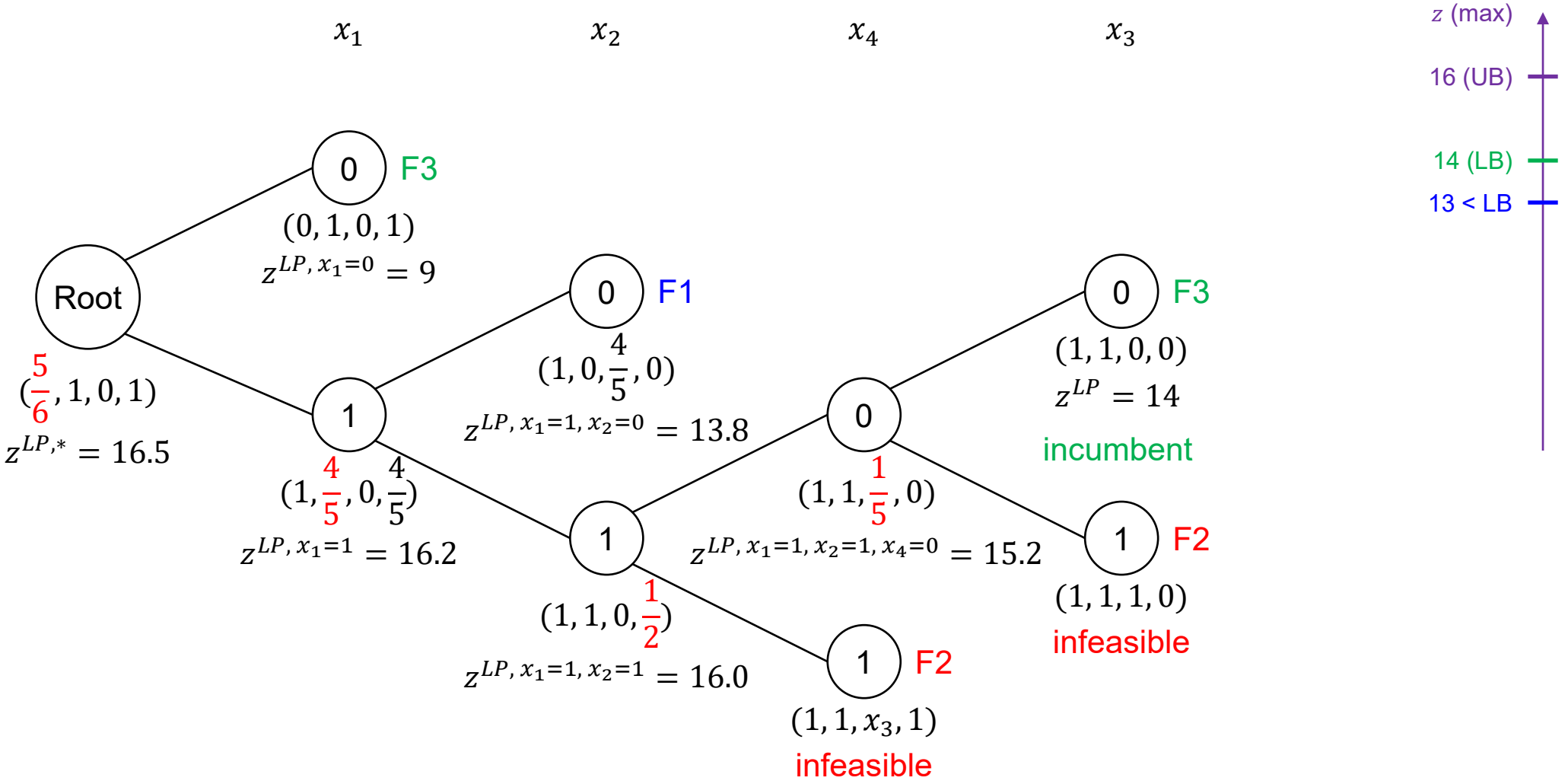

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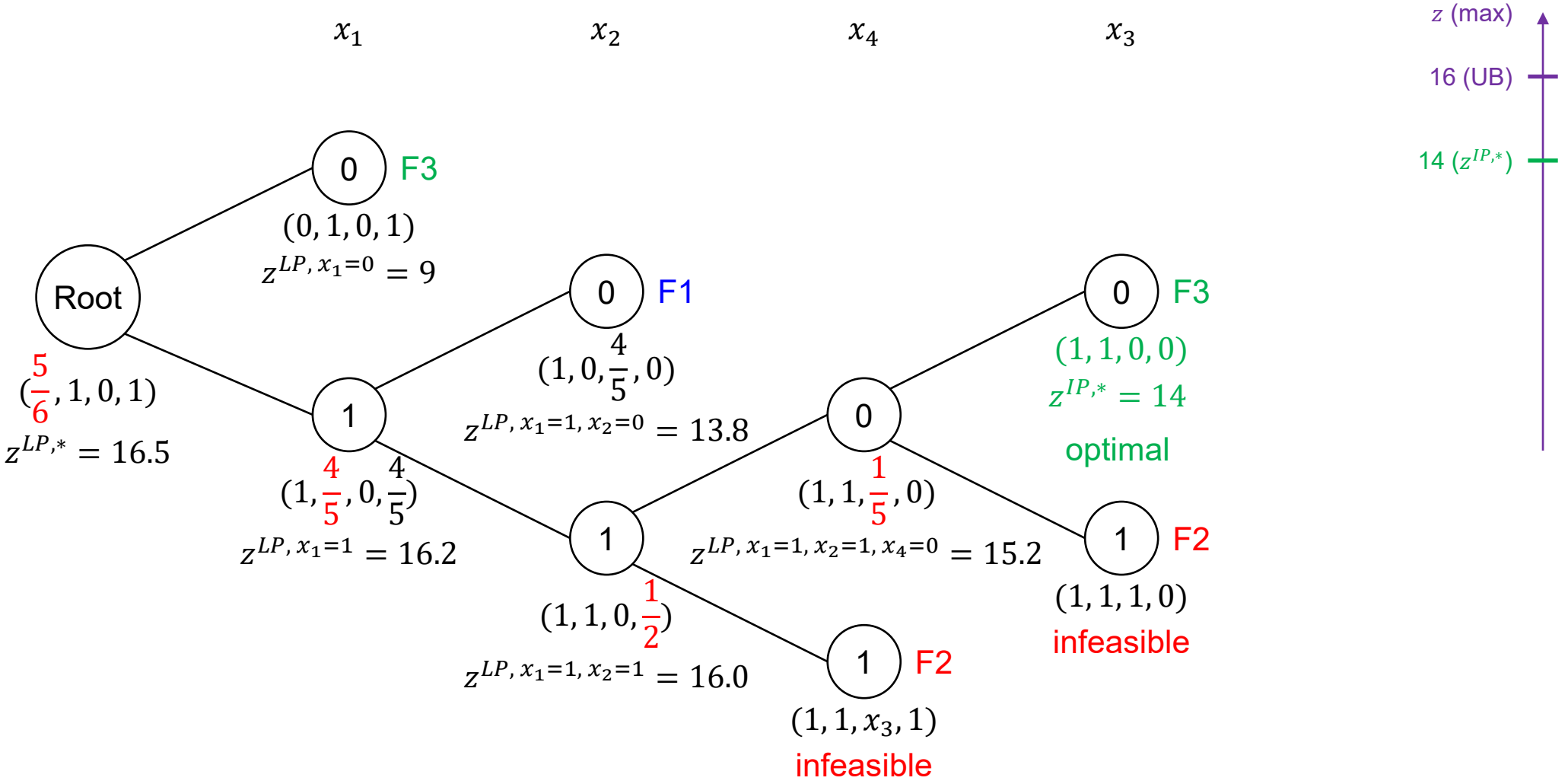
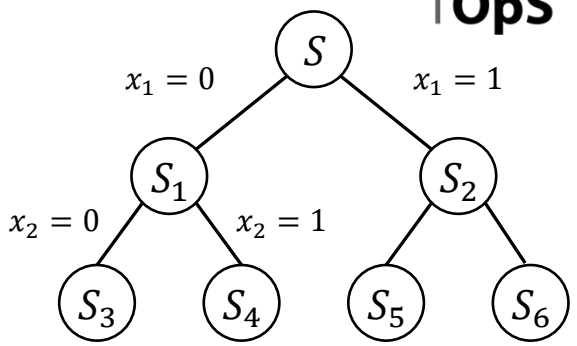


Branch and Bound

- "Implicit" enumeration
- Divide and conquer



- Branch on non-integer solutions (e.g., $x_1 = 5/6$)
  - Depth first search strategy: find a first feasible solution
  - Best-bound first search strategy: avoid branching on any node where $\bar{z}_k \leq z^{IP,*}$
- Bound (for any node k in the branch and bound tree)
  - Solve the LP relaxation $\bar{z}_k = z_k^{LP,*} = \max\{cx \mid x \in S_k\}$ ➔ upper bound on $z_k^{IP,*}$
- Fathom/Prune at node k (for maximization problems)
  - F1: Fathom/Prune by (global lower) bound: $\bar{z}_k = z_k^{LP,*} \leq \underline{z}$
  - F2: Fathom/Prune by infeasibility: no feasible solution
  - F3: Fathom/Prune by optimality: integer solution ➔ update incumbent solution and (global) lower bound ($\underline{z}$) by $z_k^{IP,*}$

*Branch and Bound

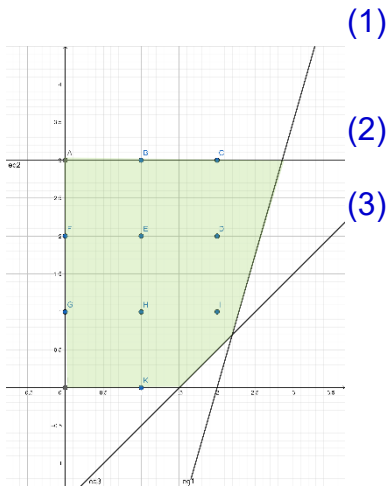
- Important questions when implementing the branch and bound method
  - Bounding: What LP relaxation or dual problem should be used to provide an upper bound (for maximation problems)?
    - Weak and “easy” vs. strong and “hard”
  - Branching: How should the feasible region be separated into smaller regions (subproblems)?
    - Fixed a priori vs. adaptive
  - Searching: In what order should the subproblems be examined?
    - Depth first search vs. Best-bound first search

➔ A combination of theory, common sense, and practical experimentation

Cutting Plane Algorithms

Cutting Plane Algorithms

- Example

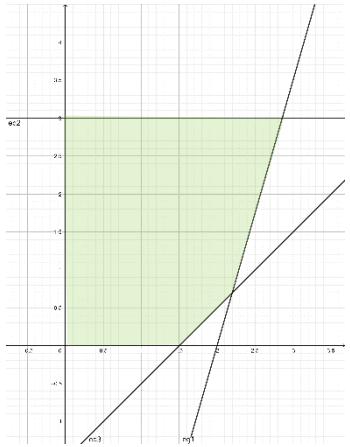


Max 4x1 - x2
s.t.
7x1 - 2x2 <= 14 (1)
x2 <= 3 (2)
2x1 - 2x2 <= 3 (3)
x1, x2 >= 0 and integer

- IP
  - Optimal solution ($x^{IP,*}$): ?
  - Optimal objective function value ($z^{IP,*}$): ?

*Cutting Plane Algorithms

- Example

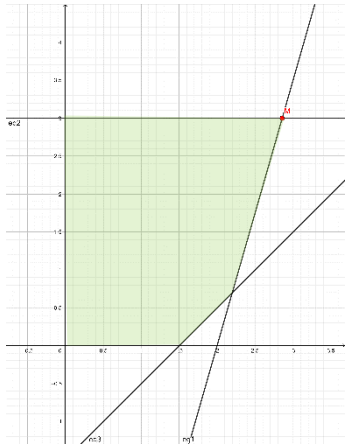


Max 4x1 - x2
s.t.
7x1 - 2x2 <= 14
x2 <= 3
2x1 - 2x2 <= 3
~~x1, x2 >= 0 and integer~~
x1, x2 >= 0

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 20/7$, $x_2 = 3$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 59/7$

*Cutting Plane Algorithms

- Example



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x2 <= 3
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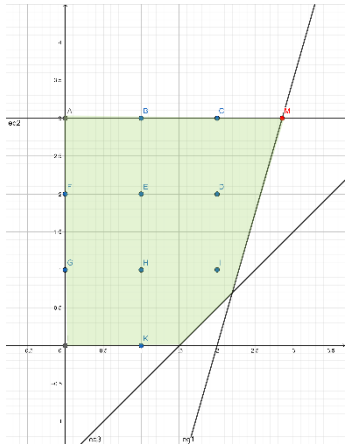
  - Optimal solution ($x^{LP,*}$): $x_1 = 20/7$, $x_2 = 3$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 59/7$

  - Question: Is $z^{LP,*} = 59/7$ an upper bound (UB) or a lower bound (LB) of $z^{IP,*}$?

*Cutting Plane Algorithms

- Example



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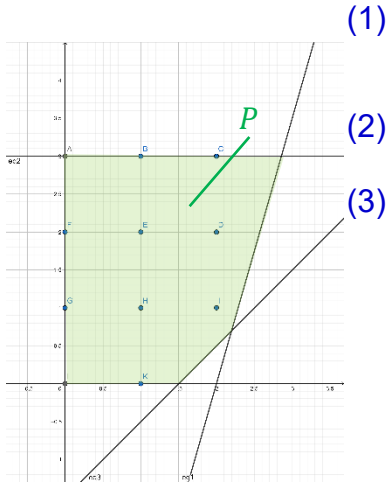
Definitions

- A polyhedron is a set of the form $\{x \in R^n | Ax \leq b\}$.
- A polyhedron $P \subset R^n$ is a formulation of a set of integral points $S \subset R^n$ if the set of integral points in P is S.



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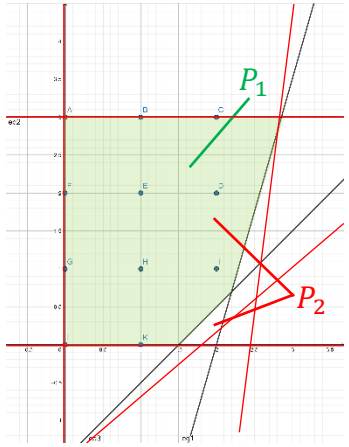
- A polyhedron $P \subset R^n$ is a formulation of a set of integral points $S \subset R^n$ if the set of integral points in P is S.
- Let S be a finite set of feasible integer solutions. Given two polyhedral formulations P_1 and P_2 for S, P_1 is a stronger formulation than P_2 if $P_1 \subset P_2$.



$P_1 \subset P_2$

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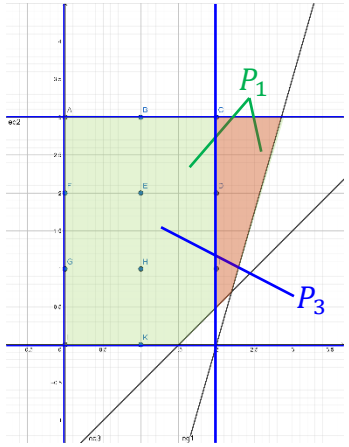

$P_3 \subset P_1$

Example: Service Centers (Recall)

- Individual (P_1)

$$
\begin{array}{lll}
\min & \sum_j f_j \sum_i d_{ij} y_{ij} & \text{minimize total distance} \\
\text{s.t.} & \sum_i x_i \leq P & \text{number of service centers} \\
& \sum_i y_{ij} = 1, \quad \forall j & \text{assign one service center to each city} \\
& y_{ij} \leq x_i, \quad \forall i, j & \text{a city can serve another city only if it is chosen to be a service center} \\
& x_i \text{ binary}, \quad \forall i & \\
& y_{ij} \text{ binary}, \quad \forall i, j &
\end{array}
$$

- Aggregate (P_2)

$$
\begin{array}{lll}
\min & \sum_j f_j \sum_i d_{ij} y_{ij} & \text{minimize total distance} \\
\text{s.t.} & \sum_i x_i \leq P & \text{number of service centers} \\
& \sum_i y_{ij} = 1, \quad \forall j & \text{assign one service center to each city} \\
& \sum_j y_{ij} \leq M x_i, \quad \forall i & \text{a city can serve other cities only if it is chosen to be a service center} \\
& x_i \text{ binary}, \quad \forall i & \\
& y_{ij} \text{ binary}, \quad \forall i, j &
\end{array}
$$

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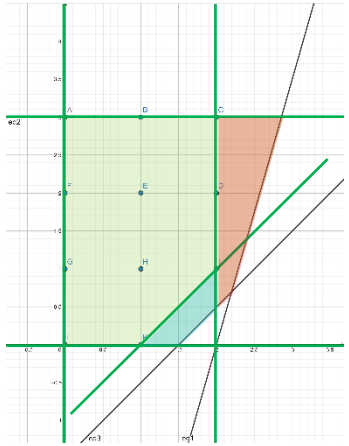
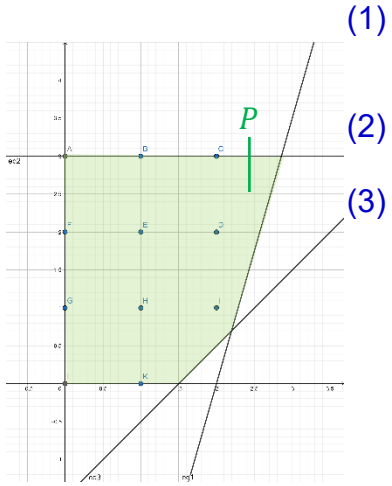
Definitions

- A polyhedron is a set of the form $\{x \in R^n | Ax \leq b\}$.
- A polyhedron $P \subset R^n$ is a formulation of a set of integral points $S \subset R^n$ if the set of integral points in P is S.
- Let S be a finite set of feasible integer solutions. Given two polyhedral formulations P_1 and P_2 for S, P_1 is a stronger formulation than P_2 if $P_1 \subset P_2$.
- A formulation is ideal if each extreme point of P is in S.



Definitions

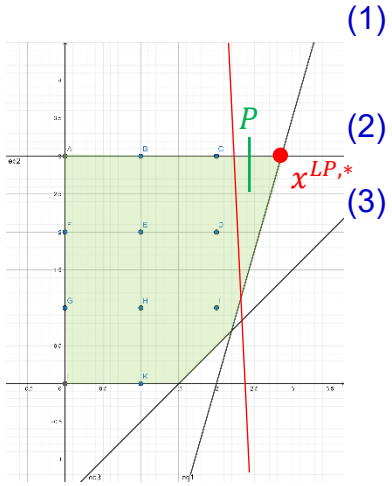
- A valid inequality for a set of integer solution S is one that is satisfied by all solutions in S.



Definitions

- A valid inequality for a set of integer solution S is one that is satisfied by all solutions in S.

- For an integer program $\max_{x \in S} c^T x$, let $x^{LP,*}$ be a noninteger optimal solution to its linear programming relaxation $\max_{x \in P} c^T x$. A cutting plane is a valid inequality for S that is not satisfied by $x^{LP,*}$.



Gomory Cutting Plane Method

- Consider the canonical form IP

$$\begin{array}{lrl} \max & z = c^T x & \\ \text{s.t.} & Ax & = b \\ & x & \geq 0, \text{integer}. \end{array}$$

*Gomory Cutting Plane Method

- Consider the canonical form IP

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0, \text{integer}. \end{aligned}$$

- Let x^* be an optimal BFS to the LP relaxation of the IP with basis matrix B. We have

$$Bx_B + Nx_N = b \qquad x_B + B^{-1}Nx_N = B^{-1}b$$

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$$\boxed{Bx_B + Nx_N = b} \qquad \boxed{x_B + B^{-1}Nx_N = B^{-1}b}$$

- If x^* is not integer, let $x_{B(i)}$ be fractional, where $x_{B(i)}$ is the ith component of x_B^*. Let $\bar{A} = B^{-1}A$ and $\bar{b} = B^{-1}b$.

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- Then the ith equation (constraint) is

$$x_{B(i)} + \sum_{j \in N} \bar{a}_{ij} x_j = \bar{b}_i$$

fractional

$x_N = 0$

*Gomory Cutting Plane Method

fractional

$$x_{B(i)} + \sum_{j \in N} \bar{a}_{ij} x_j = \bar{b}_i$$

- If the IP is feasible, this constraint is satisfied by some integral x.
- So some of the coefficients $\bar{a}_{ij}$ must also be fractional.

*Gomory Cutting Plane Method

fractional

$$x_{B(i)} + \sum_{j \in N} \bar{a}_{ij} x_j = \bar{b}_i$$

- If the IP is feasible, this constraint is satisfied by some integral x.
- So some of the coefficients $\bar{a}_{ij}$ must also be fractional.
- Since $x \geq 0$, we can round down $\bar{a}_{ij}$ and obtain the valid inequality

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \leq \bar{b}_i$$

- And then round down $\bar{b}_i$

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \leq \lfloor \bar{b}_i \rfloor.$$

- Note: It would be incorrect to round down the RHS before the LHS.

*Gomory Cutting Plane Method

fractional

$$x_{B(i)} + \sum_{j \in N} \bar{a}_{ij} x_j = \bar{b}_i$$

- If the IP is feasible, this constraint is satisfied by some integral x.
- So some of the coefficients $\bar{a}_{ij}$ must also be fractional.
- Since $x \geq 0$, we can round down $\bar{a}_{ij}$ and obtain the valid inequality

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \leq \bar{b}_i$$

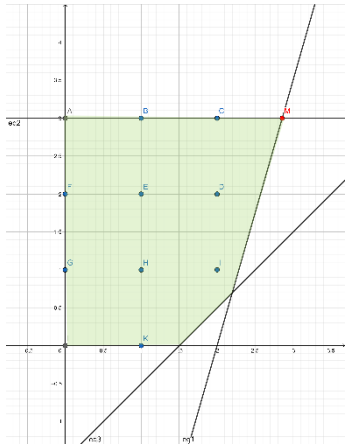
- And then round down $\bar{b}_i$

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \leq \lfloor \bar{b}_i \rfloor.$$

- Note that this inequality is violated by x^* because $\bar{b}_i$ is fractional, so that $\lfloor \bar{b}_i \rfloor < \bar{b}_i$ and $x_N = 0$.
- Thus, this is a cutting plane.

*Gomory Cutting Plane Method

- Example

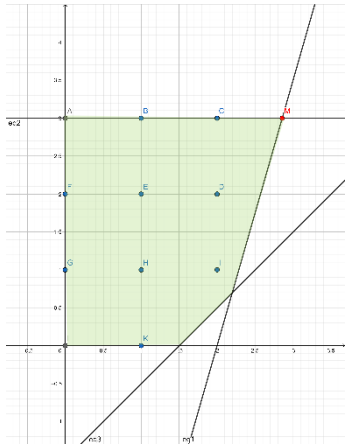


Max 4x1 - x2
s.t.
7x1 - 2x2 <= 14
x2 <= 3
2x1 - 2x2 <= 3
~~x1, x2 >= 0 and integer~~
x1, x2 >= 0

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 20/7$, $x_2 = 3$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 59/7$

*Gomory Cutting Plane Method

- Example



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  - The optimal tableau

Basis	x1	x2	s1	s2	s3	RHS
Row 0	0.0	0.0	4/7	1/7	0	8 3/7
x2	0	1	0	1	0	3
s3	0	0	- 2/7	1 3/7	1	3 2/7
x1	1	0	1/7	2/7	0	2 6/7

*Gomory Cutting Plane Method

- Gomory cutting plane

fractional

$$x_{B(i)} + \sum_{j \in N} \bar{a}_{ij} x_j = \bar{b}_i \qquad x_1 + \frac{1}{7}s_1 + \frac{2}{7}s_2 = \frac{20}{7}$$

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \le \bar{b}_i \qquad x_1 + 0s_1 + 0s_2 \le \frac{20}{7}$$

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \le \lfloor \bar{b}_i \rfloor \qquad x_1 \le 2$$

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*Gomory Cutting Plane Method

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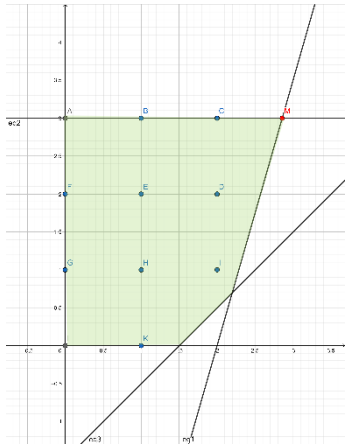

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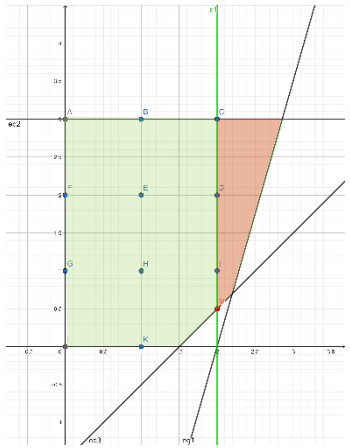
- LP with one cut
  - Optimal solution ($x^{LP,*}$): ?
  - Optimal objective function value ($z^{LP,*}$): ?

*Gomory Cutting Plane Method

- Example



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  - Optimal solution ($x^{LP,*}$): $x_1 = 2$, $x_2 = 0.5$
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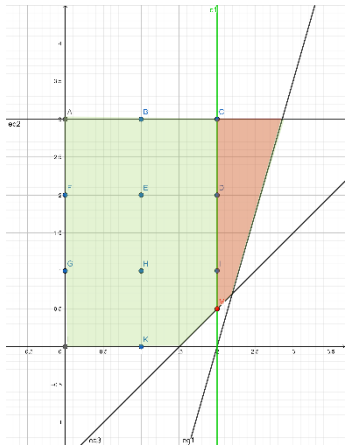
*Gomory Cutting Plane Method

- Example



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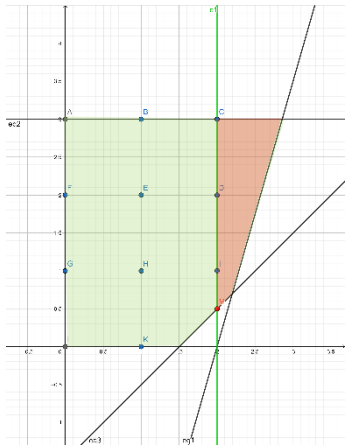
- Observe

$$\boxed{x_1 - x_2 \leq \frac{3}{2}}$$

$$\boxed{x_1 - x_2 \leq 1}$$

*Gomory Cutting Plane Method

- Example



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x1, x2 >= 0

- LP with two cuts
  - Optimal solution ($x^{LP,*}$): ?
  - Optimal objective function value ($z^{LP,*}$): ?

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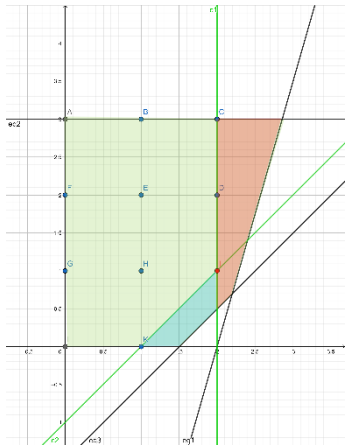
- Example



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  - Optimal solution ($x^{LP,*}$): $x_1 = 2$, $x_2 = 1$
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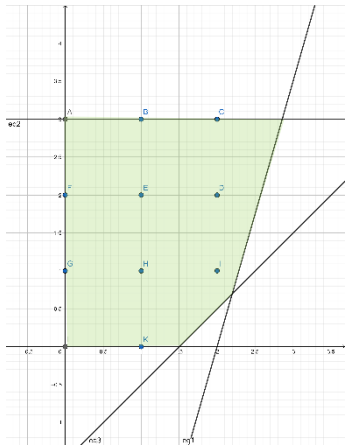

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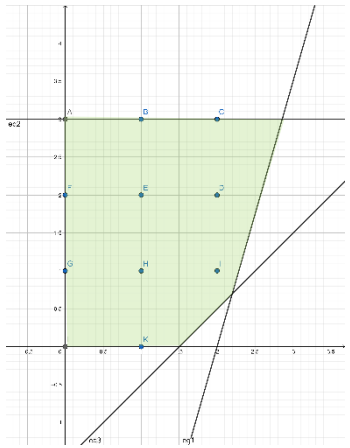

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Chvátal-Gomory (C-G) Inequality

- Consider the set S of integer solutions satisfying the constraints

$$\boxed{\begin{aligned} Ax &\leq b \\ x &\geq 0, \text{ integer} \end{aligned}}$$

and the corresponding linear relaxation P without the integer constraints.

*Chvátal-Gomory (C-G) Inequality

- Consider the set S of integer solutions satisfying the constraints

$$\boxed{\begin{aligned} Ax &\leq b \\ x &\geq 0, \text{ integer} \end{aligned}}$$

  and the corresponding linear relaxation P without the integer constraints.

- For any $u \geq 0$ and for all $x \in P$,

$$\boxed{u^T Ax \leq u^T b}$$

- Then (because $x \geq 0$),

$$\boxed{\lfloor u^T A \rfloor x \leq u^T Ax \leq u^T b}$$

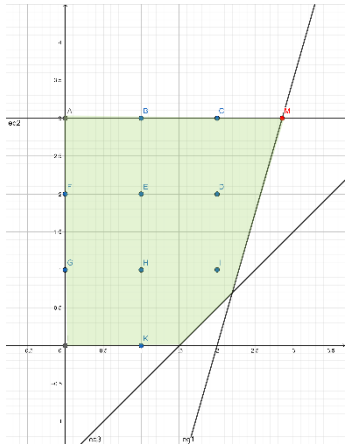
- Then for $x \in S$,

$$\boxed{\lfloor u^T A \rfloor x \leq \lfloor u^T b \rfloor}$$

- This is a valid inequality for S.

*Chvátal-Gomory (C-G) Inequality

- Example



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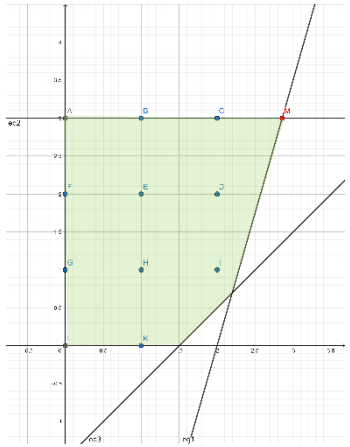
*Chvátal-Gomory (C-G) Inequality

- Example

$$Ax \le b$$
$$x \ge 0, \text{ integer}$$



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$$u = (2/7, 5/7, 0)$$

$$u^T Ax \le u^T b \qquad 2x_1 + 1/7x_2 \le 4 + 15/7$$

$$\lfloor u^T A \rfloor x \le u^T Ax \le u^T b \qquad 2x_1 + 0x_2 \le 4 + 15/7$$

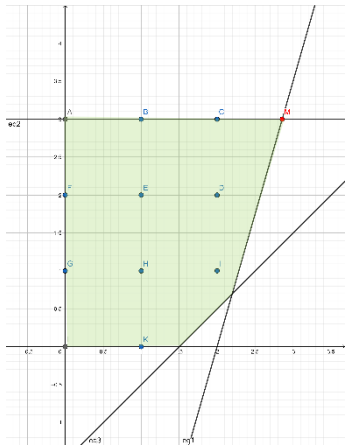
$$\lfloor u^T A \rfloor x \le \lfloor u^T b \rfloor \qquad 2x_1 \le \lfloor 4 + 15/7 \rfloor = 6$$

$$x_1 \le 3$$

*Chvátal-Gomory (C-G) Inequality

- Example

$$\begin{aligned} Ax &\le b \\ x &\ge 0, \ \text{integer} \end{aligned}$$



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$u = (2/7, 5/7, 0)$ $\qquad$ $u = (2/7, 9/14, 0)$

$u^T A x \le u^T b$ $\qquad$ $2x_1 + 1/7x_2 \le 4 + 15/7$ $\qquad$ $2x_1 + 1/14x_2 \le 4 + 27/14$

$\lfloor u^T A \rfloor x \le u^T A x \le u^T b$ $\qquad$ $2x_1 + 0x_2 \le 4 + 15/7$ $\qquad$ $2x_1 + 0x_2 \le 4 + 27/14$

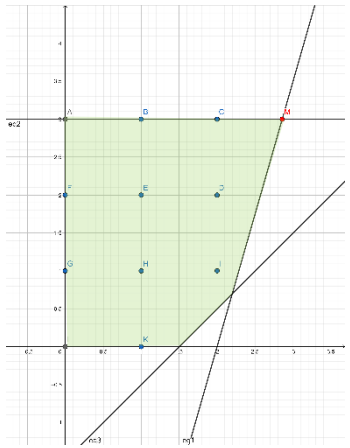
$\lfloor u^T A \rfloor x \le \lfloor u^T b \rfloor$ $\qquad$ $2x_1 \le \lfloor 4 + 15/7 \rfloor = 6$ $\qquad$ $2x_1 \le \lfloor 4 + 27/14 \rfloor = 5$

$x_1 \le 3$ $\qquad$ $x_1 \le \lfloor 5/2 \rfloor = 2$

*Chvátal-Gomory (C-G) Inequality

Example

$$\boxed{\begin{aligned} Ax &\le b \\ x &\ge 0, \ \text{integer} \end{aligned}}$$



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$u = (2/7, 5/7, 0)$ $\quad$ $u = (2/7, 9/14, 0)$ $\quad$ $u = (2/7, 4/7, 0)$

$u^T Ax \le u^T b$ $\quad$ $2x_1 + 1/7x_2 \le 4 + 15/7$ $\quad$ $2x_1 + 1/14x_2 \le 4 + 27/14$

$\lfloor u^T A \rfloor x \le u^T Ax \le u^T b$ $\quad$ $2x_1 + 0x_2 \le 4 + 15/7$ $\quad$ $2x_1 + 0x_2 \le 4 + 27/14$ $\quad$ $2x_1 + 0x_2 \le 4 + 12/7$

$\lfloor u^T A \rfloor x \le \lfloor u^T b \rfloor$ $\quad$ $2x_1 \le \lfloor 4 + 15/7 \rfloor = 6$ $\quad$ $2x_1 \le \lfloor 4 + 27/14 \rfloor = 5$ $\quad$ $2x_1 \le \lfloor 4 + 12/7 \rfloor = 5$

$x_1 \le 3$ $\quad$ $x_1 \le \lfloor 5/2 \rfloor = 2$ $\quad$ $x_1 \le \lfloor 5/2 \rfloor = 2$

*Chvátal-Gomory (C-G) Inequality

- Chvátal-Gomory inequality is a very general approach (for any $u \geq 0$).

- Gomory cuts are a special case of Chvátal-Gomory inequalities where the constraint multipliers u are a row of B^{-1}.

Theorem: Every valid inequality for $S = P \cap Z^n$, where $P = \{Ax \leq b, x \geq 0\}$, can be obtained by applying the Chvátal-Gomory procedure a finite number of times.

*Recap

Integer Program (IP)

- Finite number of feasible solutions
  - Exponential growth!!!
- Optimal IP solutions are NOT necessarily CPF/BF's.
- Measures of difficulties
  - Number of integer variables
  - Binary or general integer
  - Special structures
  - ……

*Round to Integer?

- Example

Max 9x1 + 5x2 + 6x3 + 4x4
s.t.

$$
\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
-x_1 + x_3 &\le 0 \\
-x_2 + x_4 &\le 0 \\
\cancel{x_1, x_2, x_3, x_4 \text{ binary}} & \\
0 \le x_1, x_2, x_3, x_4 &\le 1
\end{aligned}
$$

- LP
  - Optimal solution ($x^{LP,*}$): $x_1 = 5/6$, $x_2 = 1$, $x_3 = 0$, $x_4 = 1$
  - Optimal objective function value ($z^{LP,*}$): $z^{LP,*} = 16.5$

  - $x_1 = 0$ ➔ Objective function value $z^{(0,1,0,1)} = 9$ ➔ LB
  - $x_1 = 1$? ➔ Infeasible!

*How to Solve an IP? (Recap)

- Example

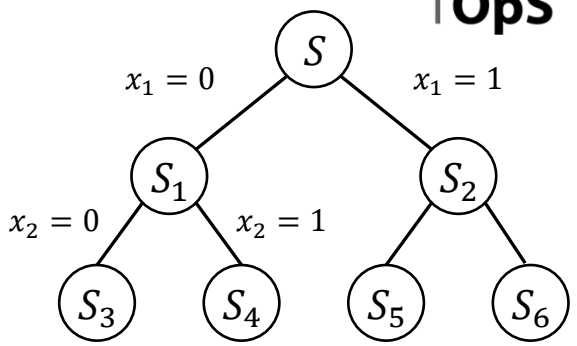
$$
\begin{aligned}
\text{Max } & 9x_1 + 5x_2 + 6x_3 + 4x_4 \\
\text{s.t. } & \\
& 6x_1 + 3x_2 + 5x_3 + 2x_4 \le 10 \\
& x_3 + x_4 \le 1 \\
& -x_1 + x_3 \le 0 \\
& -x_2 + x_4 \le 0 \\
& x_1, x_2, x_3, x_4 \text{ binary}
\end{aligned}
$$

- IP
  - Optimal solution ($x^{IP,*}$)?
  - Optimal objective function value ($z^{IP,*}$)?

  - LP relaxation: $z^{LP,*}$ $(= 16.5)$ is an upper bound (UB) of $z^{IP,*}$.
  - Feasible integer solution: $z^{(0,1,0,1)}$ $(= 9)$ is a lower bound (LB) of $z^{IP,*}$.

Branch and Bound

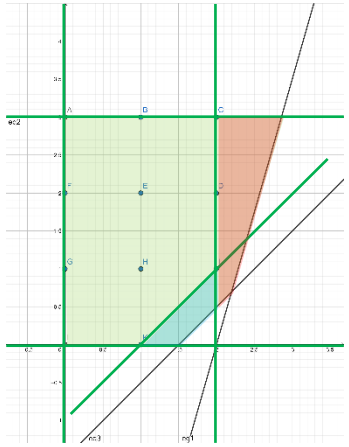
- "Implicit" enumeration
- Divide and conquer



- Branch on non-integer solutions (e.g., $x_1 = 5/6$)
  - Depth first search strategy: find a first feasible solution
  - Best-bound first search strategy: avoid branching on any node where $\bar{z}_k \leq z^{IP,*}$
- Bound (for any node k in the branch and bound tree)
  - Solve the LP relaxation $\bar{z}_k = z_k^{LP,*} = \max\{cx \mid x \in S_k\}$ ➔ upper bound on $z_k^{IP,*}$
- Fathom/Prune at node k (for maximization problems)
  - F1: Fathom/Prune by (global lower) bound: $\bar{z}_k = z_k^{LP,*} \leq \underline{z}$
  - F2: Fathom/Prune by infeasibility: no feasible solution
  - F3: Fathom/Prune by optimality: integer solution ➔ update incumbent solution and (global) lower bound ($\underline{z}$) by $z_k^{IP,*}$

Definitions

- A polyhedron is a set of the form $\{x \in R^n | Ax \leq b\}$.
- A polyhedron $P \subset R^n$ is a formulation of a set of integral points $S \subset R^n$ if the set of integral points in P is S.
- Let S be a finite set of feasible integer solutions. Given two polyhedral formulations P_1 and P_2 for S, P_1 is a stronger formulation than P_2 if $P_1 \subset P_2$.
- A formulation is ideal if each extreme point of P is in S.



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- So some of the coefficients $\bar{a}_{ij}$ must also be fractional.
- Since $x \geq 0$, we can round down $\bar{a}_{ij}$ and obtain the valid inequality

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \leq \bar{b}_i$$

- And then round down $\bar{b}_i$

$$x_{B(i)} + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor x_j \leq \lfloor \bar{b}_i \rfloor.$$

- Note that this inequality is violated by x^* because $\bar{b}_i$ is fractional, so that $\lfloor \bar{b}_i \rfloor < \bar{b}_i$ and $x_N = 0$.
- Thus, this is a cutting plane.

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$$\boxed{u^T Ax \leq u^T b}$$

- Then (because $x \geq 0$),

$$\boxed{\lfloor u^T A \rfloor x \leq u^T Ax \leq u^T b}$$

- Then for $x \in S$,

$$\boxed{\lfloor u^T A \rfloor x \leq \lfloor u^T b \rfloor}$$

- This is a valid inequality for S.

Q & A